

SCIENCE

INSTANT LESSONS POWERPOINT

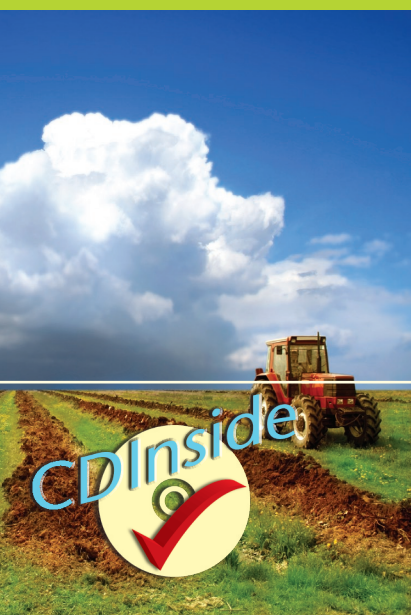
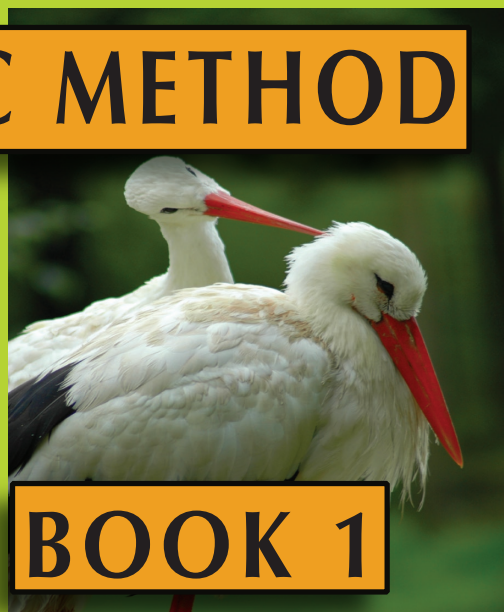
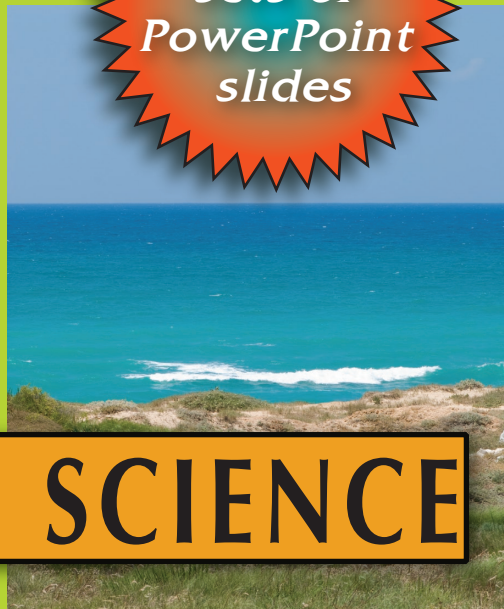
20
sets of
PowerPoint
slides

THE SKILLS OF SCIENCE

SCIENTIFIC METHOD

BOOK 1

Donna Bennett



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BOOK 1

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This book was commissioned by Jennifer Gough, project management was by Piper Print Media, Suellen Entwisle designed the cover and the PowerPoint presentations, and David Dickson supplied the illustrations.



Donna Bennett is a highly qualified and experienced teacher. She is the winner of two national awards, the CRA Mining Company Fellowship and the BHP Science Teacher Award, for innovative and creative delivery of science. She is the Director of Science Education Australia and her science workshop programs are always popular with teachers and students alike. Donna has written several successful science education books, established the Queensland Science Marathon and has taken her 'Science for Bush Kids' program to regional areas. She has travelled internationally where her special form of science education has been in demand.

Donna believes that science is the greatest adventure that has ever been devised. She sees teachers as consultants and cheerleaders who provide probing questions, enthusiasm and confidence and who tease out an understanding of the basic notions of science. Teachers are a source of knowledge and information, which is shared in order to guide student exploration, rather than define it. As an author she has a clear and effective style and believes that interactive science investigations contribute most to learning when they are part of a well-thought out plan that triggers the natural curiosity of students. She provides a freedom that allows students the pleasure of building their own knowledge and understanding. Once they get over the shock of having this type of responsibility, and experience the thrill of succeeding, nothing can stop them.

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The scientific method is simple and makes really good sense. Some people make a lifetime career out of defining scientific method – others just use it!

Science is like a game about understanding the universe and this game is played by special rules called the 'scientific method'. The game usually involves asking a question, gathering information about the question, forming the hypothesis, putting it to the test and telling others about what you found out. When people play this game, they must remember not only that the rules are flexible and provide only a general plan, but that it usually leaves out the important bits. Anyone who loves science knows that these missing bits, such as hunches, wrong turns, intuition and the ideas that didn't work make scientific discovery the most challenging, exhilarating, frustrating, yet exciting adventure ever devised.

This series is about the nature of science and testing ideas. It encourages an everyday perception of science, conceptual understanding and thinking skills.

The activities:

- encourage students to learn how to think and how to explore scientifically
- encourage the ability to see a good question, investigate it and draw evidence-based conclusions
- introduce the notion of a fair comparison as a precursor to the idea of 'controlled experiments'
- emphasise that in science, nothing can be proved – only disproved!
- will instill a confidence in conversing about science and questioning claims about scientific matters
- develop the ability to make informed decisions about global and everyday issues.
- separate science from myth, considered opinion from hasty judgment and will help students to realise that of all the things that are scientifically possible, not all are desirable.

It is worth remembering what Galileo Galilei (1546-1642), the 'father' of scientific method said: *The experiment is the ordeal through which all ideas must pass before they are accepted as truth.* The game of science is about curiosity, which turns into the satisfaction of knowledge – it should begin with some fun and end with wisdom.

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What does it include?

The topics in this title are complemented by the information and activities contained on the CD-ROM at the back of this book.

The CD-ROM includes:

- 20 PowerPoint presentations to accompany each of the 20 topics in this book, which summarise and extend the content.
- 20 follow-up activity sheets, each based on the matching PowerPoint presentation. These are supplied as PDF documents.

Three video-clips are also provided on the CD-ROM, as extension material for the topics listed below.

Topic 1 – Observation and asking scientific questions, in the context of water drops

Topic 2 – The hierarchy of classification, using the horse and the zebra as examples.

Topic 19 – Ethics in science: using animals to research medical conditions such as Parkinson's Disease.

Note to the teacher

To get the most out of these teaching resources, teachers are encouraged to preview the Powerpoint presentations and accompanying activity sheets before using them in class.

Using the CD-ROM

In the Windows environment, the CD-ROM will autorun. From the Main menu, simply click the PowerPoint presentation you wish to view.

In the Macintosh environment, double-click the file index.pps. From the Main menu, click the PowerPoint presentation you wish to view.

You can also open each presentation from within the PowerPoint application.

The Slide Show view is used to display each PowerPoint presentation on the screen. It is recommended that you use an external projector for whole-class use.

To access the activity sheets, simply click the pencil icon on the appropriate topic.

To access the video-clips, navigate through Windows Explorer (for Windows users) or Finder (for Macintosh users) of the CD-ROM, and open the folder 'Videos'. Double-click the video-clip you wish to play.

Technical specifications

Windows

- The CD-ROM will autorun in the Windows environment, with an operating system of Windows 97 or later.
- To edit a presentation, open it from within the PowerPoint application. Note that if editing these PowerPoint presentations in a version prior to Office 2000, cosmetic losses will occur in some animations, and visual effects such as drop shadows may be lost. These losses are cosmetic only, and will not affect the clarity and content of the slide show.

Macintosh

- Double-click the file index.pps on the CD-ROM. From the Main menu, click on the presentation you wish to view.
- To view a fully featured presentation, and edit it, you must have the PowerPoint 2004 application installed. (i.e. Office 2004 for Macintosh or later)
- To edit a presentation, open it from within the PowerPoint application. Note that if editing these PowerPoint presentations in a Macintosh environment, cosmetic losses will occur in some animations, and visual effects such as drop shadows may be lost. These losses are cosmetic only, and will not affect the clarity and content of the slide show.

Where do questions come from?

Observing starts people asking questions about something that interests them. Observing things involves using our senses, paying attention and analysing situations in great detail. Scientists are always observing, exploring, examining the world and asking questions. Science is what scientists use to answer their questions.

Wondering or open questions

Wondering questions can start scientists on their search for answers. Leonardo da Vinci wondered about how things fly. Charles Darwin was curious about the origin of animal life on the planet. Rachel Carson wondered how pesticides affected the environment. There are different kinds of questions, depending on what people want to do or want to find out.

Activity 1

- 1.1 The universe is full of things to wonder about. What kinds of questions do you wonder about? Things to do with you, such as why you have hair in your nose? Things about long ago, such as how scientists can tell what a dinosaur looked like from a single bone? Things about far away, such as how do scientists measure the size of the universe? List two things you really want to know about:
- a your body
 - b the past
 - c space
 - d another topic or area.

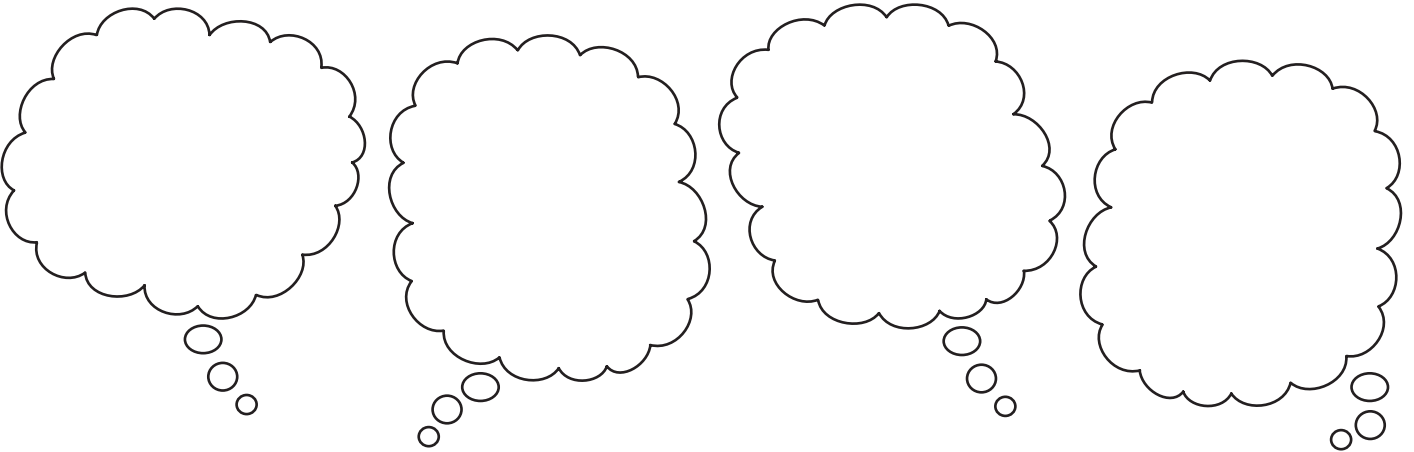
Open and closed questions

Scientists tend to ponder *open* questions which often have many acceptable answers. For example: *What do you think life on Earth would be like if the atmosphere contained twice as much oxygen as it does now?* To find an answer to this huge question, scientists look for answers to lots of smaller *closed* questions, then put all the answers together. Closed questions generally have a limited number of answers or even yes or no answers. They give you facts and are quick and easy to answer.

Activity 2

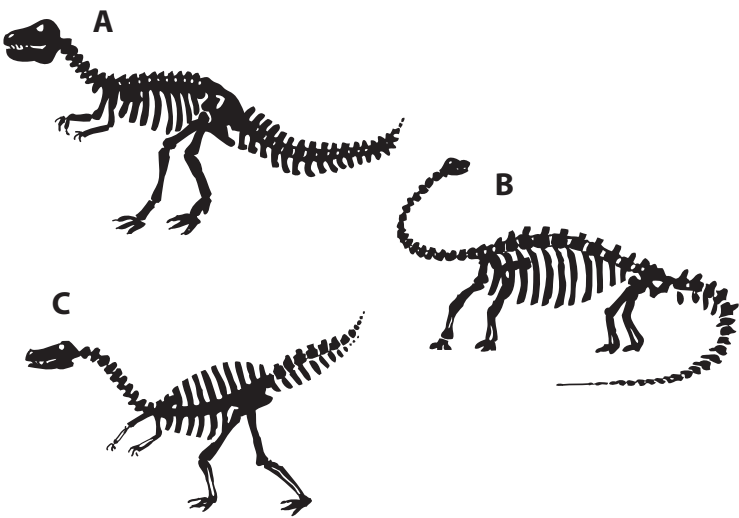
- 2.1 In the table at the right, identify each question as 'open' or 'closed' by filling in the last column.
- 2.2 Use the thought bubbles to write four open questions that you think scientists today are wondering about and working on.

	QUESTION	OPEN OR CLOSED?
a	What time will we start?	
b	Do you agree with this theory?	
c	What is an example of a flower?	
d	What is the definition of an atom?	
e	What predictions can you make based on the available data?	



2.3 The ability to ask good closed questions is an important skill for a scientist. It can help people to identify a particular object from among a group of similar objects.

- a** Scientists have learned about dinosaurs from fossilised skeletal remains, footprints, droppings and eggs. Study the skeletons at the right.
- b** Choose one skeleton: A, B or C.
- c** Write a series of four closed questions about your skeleton that would help someone work out which skeleton you chose.



Action questions

Action questions begin with words such as:

- What ... ?
- Will ... ?
- How ... ?
- Do ... ?
- When ... ?
- Have ... ?

When answering an action question, scientists first guess what the answer is and then do a test to find out if their guess was right. Take the English doctor Edward Jenner, for example. In his time, many people died of a disease called small pox. His action question may have been: *What will happen if a person who has had cow pox is infected with small pox?* (Cow pox was a disease that people did not die from.) He guessed that having cow pox would stop people from dying of small pox. He tested this and obtained evidence that vaccinating people with cow pox prevented them from being infected with small pox. Jenner’s work resulted in the eradication of small pox.

Activity 3

- 3.1** Close your eyes and think about popcorn for one minute.
- 3.2** Using the words in the table below, write four more action questions about popcorn.

	WORD	ACTION QUESTION
a	Does	<i>Does all corn pop?</i>
b	Why	
c	Will	
d	How	
e	What	

Try not to fool yourself!

Science is about answering questions that can be quite puzzling and may take a long time to figure out. It took 400 years to find answers to all the questions that finally enabled people to fly around the world in fixed winged aircraft.

Activity 4

Work out answers to the following questions.

- 4.1** Peter, Sarah and Jake shared a house. One day Rosie came to live with them. Peter and Sarah went out to a party that evening. When they returned, Jake was hovering over Rosie’s dead body. The police did not arrest Jake. He wasn’t even questioned about the crime. Why?
- 4.2** A man was out jogging when he felt the first raindrops fall. He didn’t have an umbrella or a raincoat or a hat. He started to jog faster and the rain got heavier. The rain wet his clothing and his running shoes and it dripped off his nose. However, his hair did not get wet. Why?

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM!

What makes an elephant an elephant, a carrot a carrot or a rock a rock? As scientists explore the world, they love to sort everything they find out about into groups, according to its similarities and differences. This is called **classification**. Classification requires careful observation and meticulous attention to detail, and involves comparing and describing.

Why classify?

To classify objects is to arrange them into groups. How do you classify your clothing? Dirty on the floor and clean in the ironing basket, some shoes under the bed and others in the cupboard, socks jumbled together or matched into pairs based on colour?

Activity 1

1.1 Human beings are all similar, but no two people are exactly alike, except identical twins! There are many variations in the basic human plan. Some variations are internal, like blood type, and others are obvious, like your nose shape and hair colour. Each person has their own special characteristics. Take Sam, for example. He has eyes with short lashes, blonde hair and dimples. He can raise one eyebrow at a time but he cannot roll his tongue into a lengthwise tube. Sam’s mum has blonde hair and long lashes but no dimples. Sam’s dad has brown hair, short lashes and dimples. Both parents can roll their tongue but only Sam’s dad can raise one eyebrow.

Using the columns in the table below, tick the traits that Sam inherited from his father and those that he got from his mother.

TRAIT	SAM’S CHARACTERISTICS	SAM’S MOTHER	SAM’S FATHER
Hair colour	Light		
Dimples	Yes		
Eye lashes	Short		
Ability to roll tongue into a lengthwise tube	No		
Ability to raise one eyebrow only	Yes		

1.2 List five of your own special characteristics.

Take the easy way

Classifying objects like rocks is a stepping stone to classifying more complex living things. Plants and animals can be quite difficult to classify. Remember that detailed observation is essential.

Activity 2

2.1 What makes a rock a rock? What makes one rock different from another rock? Rocks are first classified according to the way they were formed. The size and arrangement of minerals within them also helps to identify rocks. The table below contains descriptions of how rocks on Earth are formed. Study the five rock diagrams on the next page and write the number of each rock in the correct row of the table.

GROUP	DESCRIPTION	ROCK NUMBER
A	These rocks have formed from cooled and hardened magma from deep inside the Earth. They are mostly very hard and often contain large, shiny crystals.	
B	These rocks are formed by bits of other rocks that are pressed together in layers or ‘glued’ together in bunches. They may contain fossils.	
C	These rocks have been changed by very high temperatures and squeezed by extreme pressures. The crystals may be distorted and layers may be twisted.	



Rock 1



Rock 2



Rock 3



Rock 4



Rock 5

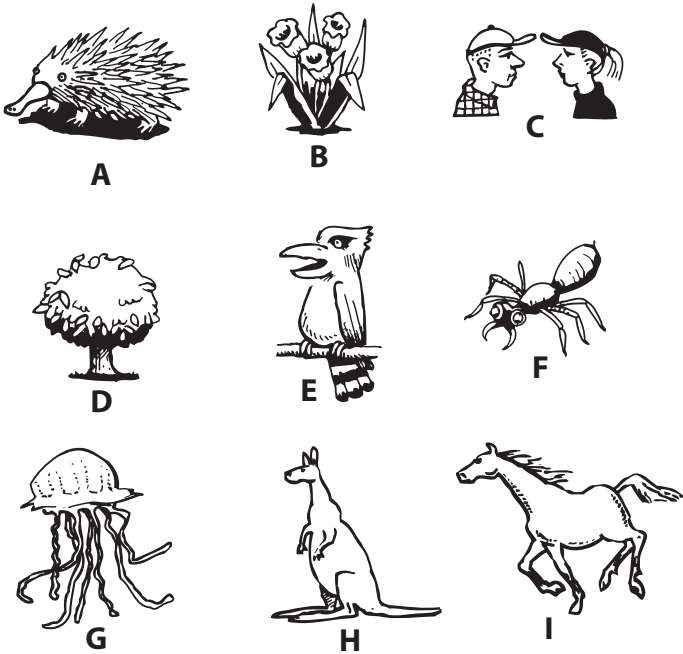
Take the hard way

Most living things fall into two main groups – plants and animals. What is the difference between a plant and an animal? Plants usually stay in one place, make their own food, do not have sensory organs, are usually green and have a cellulose ‘skeleton’. Animals move around, have a boney skeleton, have a sensory system, eat other organisms for food and are many other colours as well as green.

Activity 3

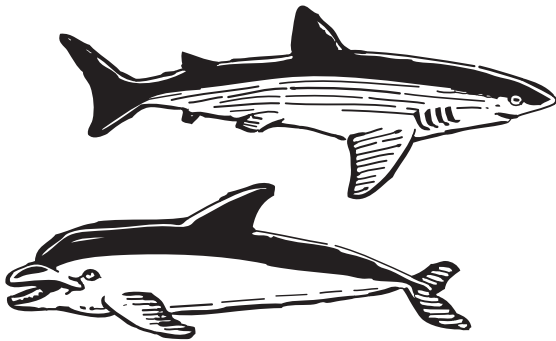
3.1 There are many ways to sort living things. Group the animals and plants in the illustration into the categories shown in the table.

CATEGORY	GROUP
Plants	
Animals that live in water	
Animals that do not live in water	
Animals with bones	
Animals without bones	
Animals that are eaten by other animals	
Animals that are not eaten by other animals	



3.2 When you classify things you must look beyond what you can actually see. For example, it is easy to see that dolphins and sharks have both become streamlined so that they can catch food. However, scientists classify sharks as ‘fish’ and dolphins as ‘mammals’.

Study the diagrams and list two differences between sharks and dolphins that are easily seen and two that are not easily seen.



Putting a name to it

When a life form is classified, it is usually given a two-part scientific name that identifies it and shows which group it belongs to. This is called a **binomial name** and consists firstly of a genus name and secondly a species name. For example, the scientific name for the grey kangaroo is *Macropus giganteus*. Using these Latin names instead of common names allows scientists to communicate about a particular living organism, no matter which language they normally speak. It is the agreed international language of identification.

Activity 4

4.1 The name *Macropus rufus* is the universally accepted name for the red kangaroo. Write the correct part of the name in the table below.

GENUS NAME	SPECIES NAME	COMMON NAME

4.2 Why do scientists love to arrange everything into ordered systems?

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM!

The scientific process

Intuition and inspiration are an important part of science. They are part of the systematic procedure called the scientific process which scientists use to find answers.

Activity 1

1.1 Humans spend a lot of time solving the problems they see in the world around them.

- a** What problem solving do you do in order to get to school on time?
- b** How could you solve the problem of completing your homework when you have left your book at school?

1.2 Scientists spend a lot of time solving problems. What information do you think they would consider in order to:

- a** solve water shortages in Australia?
- b** reduce greenhouse gas levels in the atmosphere?

Using what you know

When you try to solve a problem:

- clarify the problem or the question that you want to solve or answer.
- try to remember things that you know or can do that might help you.

The problem must be a question that can be answered by some sort of investigation.

Activity 2

2.1 Imagine that you have tried to turn on a torch, but nothing happens.

- a** From this observation, what problem have you identified?
- b** Based on your knowledge of torches and how they work, list three likely reasons for why the torch is not working.

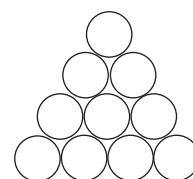
2.2 Suppose that you have spilled the salt and the pepper and they are mixed up together. To separate the salt from the pepper, you could pick all the pepper out by hand – but this would be slow and boring. Use your knowledge of science to solve the problem of separating the salt from the pepper.

Blind leaps and other strategies

When scientists have identified a problem, sometimes they just jump in and start trying things without thinking through a sensible method. This strategy can become very frustrating. One alternative is to approach the problem from different directions.

Activity 3

3.1 Ten circles have been arranged in the shape of a triangle, as shown at the right. Which three circles must you move in order to make the triangle point in the opposite direction? Use a trial and error approach to solve this problem.



3.2 Scientists sometimes work backwards from where they want to end up. Use this strategy to solve the problem of the circles.

Problem solving often means trying more than one strategy before you find one that works.

Activity 4

4.1 A tennis player knows that she will have an advantage over her opponent if she can get the ball to spin back towards her after it passes over the net. She tries the strategies listed on the next page. Which one do you think will be the most successful? Why?



- a** Hit the ball hard and downwards.
- b** Hit the ball from underneath and slowly.
- c** Spin the racquet hard over the top of the ball with a downward movement.
- d** Hit the ball hard and fast from the back of the court.

Solving problems can involve guessing what will happen if you do a certain thing. When you guess and then test the guess, sometimes things don't happen the way you thought they would. Then you have to figure out why they didn't happen as you anticipated.

Activity 5

5.1 A teacher showed a group of students how to make 'rubber eggs'. They removed the shell from a raw egg by soaking it in vinegar and guessed that it would bounce when dropped on the floor. The students were surprised to find that the egg splattered all over the floor.

- a** Was their guess correct?
- b** Why didn't the egg bounce?

The building blocks of problem solving

Knowing and understanding how things work often helps to solve a problem. The Bernoulli Principle says that as the speed of air (or liquid) increases, its pressure drops, and as its speed decreases, its pressure increases.

Activity 6

6.1 Imagine two ping pong balls, attached to two pieces of cotton, hanging side-by-side about 2 cm apart. If you understand the Bernoulli Principle of moving air, making the two ping pong balls move towards each other without touching either of them is a simple problem.

- a** How would you solve this problem?
- b** What was it that you understood from the Bernoulli Principle and applied to this problem?

6.2 Kites were flown in China more than 2000 years ago, long before anyone knew how they worked. How would your understanding of the Bernoulli Principle assist you in designing a kite?

6.3 In the 16th century, Leonardo da Vinci studied the flight of birds but all his attempts to build flying machines failed. However, he did have enough understanding of flight to design a circular flying machine, which is now called a helicopter. His solution to the problem of flying waited 400 years, until the petrol engine was invented. Why did this grand idea about flying have to wait for another major invention?



Working with what you have

Sometimes, problem solving involves recognising limitations and risks. The real challenge is to devise solutions that give good results in spite of the restrictions.

Activity 7

7.1 Your task is to make a bag for carrying two kilograms of potatoes.

- a** Identify the limitations of using one sheet of newspaper and some sticky tape for this task.
- b** What risk is involved if the bag fails?
- c** Considering the limitations of the materials, suggest two ways of making the bag strong enough for the task.



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM!

What is research?

Research is an activity that is based on the work of others. It is the process through which new knowledge is discovered. The word *research* means 'to investigate thoroughly'. There are different forms of research.

Science in action – designing sports shoes

Sports shoes have become highly sophisticated. They are designed not only for exceptional sporting performance, but also for their aesthetic appeal in order to maximise commercial sales.

To protect the legs and feet from injury, sports shoes can contain air, gel cushioning, air pumps and energy return soles. They are largely constructed of thermoplastics, and successful designs allow air in and smelly odours out.

Research – the first step

Scientists begin with a literature search. A literature search involves reviewing currently available information or data that relates to the proposed research area. This does not mean that scientists just copy the work of others – rather, they search through the work that has already been done. This helps them define exactly what information is missing from the known body of knowledge as well as suggesting the best way to proceed with the investigation.



Activity 1

- 1.1** A concept map is a method of exploring links between a number of ideas or concepts on a particular topic. The map becomes a summary of a person's current understanding on a particular theme. Use the following steps to make your own concept map.
 - a** Brainstorm everything that you know about athletic shoes – for example, *polymer, foot support, flexibility, smell, comfort, colour*. Write down each word.
 - b** Think of any connections between the words you listed. Use lines to link the words and write a short explanation of the connection between them along the connecting line.
 - c** Draw blank circles where there are gaps in your knowledge or where you need to find out more.
 - d** Highlight the key words and ideas.
- 1.2** Once they have identified key terms and concepts, scientists use reference books and journals to expand and broaden their knowledge of the topic. Use sports books and magazines to find at least two articles about sports shoes.
- 1.3** The Internet enables researchers to confer regularly with scientists around the world about new discoveries that may contribute to their own work. Using your key terms and the Internet as a research tools, find out more about the latest developments in the sole construction of sports shoes.
- 1.4** The type of research you have been conducting is the part where you gather information about the question. Scientists track their research by recording their readings, information evaluations and source details as they go. Sort the information that you have found into a sensible order and compile a reference list.

Pure research and applied research

Pure research is exploratory research, often driven by the researcher's curiosity, interest or even a hunch. It may not have any expected results in particular but it usually provides the foundation for further research. It can be difficult to obtain funding for pure research because the results may have no immediate commercial application. Pure research is traditionally followed by applied research, which usually has some practical applications in mind.

Activity 2

2.1 For each piece of research information, decide whether it is pure research or applied research and tick the appropriate column. Then decide if the research information would be directly useful in designing sports shoes, and write 'yes' or 'no' in the last column to record your decision.

RESEARCH INFORMATION		PURE RESEARCH	APPLIED RESEARCH	USEFUL INFORMATION?
a	Carbon and hydrogen are the most common units in polymers.			
b	The physical properties of a polymer can be changed by adding other materials. This is called 'compounding'.			
c	Thermosetting plastics are used in sports shoes as they can be heated and cooled many times without being ruined.			
d	Polymers have been tested for how they feel against the skin.			
e	The more fluid a polymer is, the faster its surface will recover when deformed, but the less durable it is.			
f	Isaac Newton found that for every force (action) there is an equal and opposite force (reaction).			
g	Elastomers are a type of plastic that can regain their structures after stretching.			
h	Some polymers are stronger than others.			
i	Softer plastics (ones that flow easily) are less suitable for that part of the sole under the heel of the foot.			
j	Adding water to a polymer makes it more pliable.			
k	Adding talc powder to a polymer makes it more bouncy.			
l	Elastollan can be dyed any colour at all.			
m	Air can move through Elastollan.			
n	The structure of a human foot.			
o	A plastic that bounces is not necessarily a good shock absorber.			

Activity 3

3.1 Use the information in activity 2 to help you answer the following questions.

- a Where in the sole of an sports shoe would you put a polymer with the fastest flow rate (the most fluid)?
- b Where in a sports shoe might a strong polymer be most appropriate?
- c Where in a sports shoe would you put a material with elastic or bouncy characteristics?
- d How can durability or resistance to wear be taken into consideration in the design of sports shoes?
- e Which piece of research information in the table above is not important to the sole design of a sports shoe?



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM!

Getting the guess right

Scientific researchers usually start with a guess when trying to explain natural phenomena that they have observed. However, they always carefully test the guess for accuracy by conducting experiments. Scientists call the guess a *hypothesis*. A hypothesis is a possible explanation or answer for something that has been observed. It should have the capacity to result in a 'yes', 'accepted' or 'no, 'unaccepted' answer when the experiment has been completed.

Sack the suntan

The sun radiates energy of different wavelengths, including ultraviolet, visible and infra-red radiation. Our eyes can only detect those wavelengths that are in the visible part of the spectrum – red, orange, yellow, green, blue, indigo and violet. Ultraviolet (UV) radiation is invisible. Scientists believe that UV radiation causes damage to human skin and skin cancer has been linked to UV radiation.

Activity 1

- 1.1** A group of friends spent the day at the beach, surfing. They were all wearing board shorts and T-shirts. When they returned home, they noticed that they were sunburned not only on the exposed parts of their bodies but also, to a lesser degree, under the T-shirts. This observation intrigued the friends, who thought that you couldn't get burned if you were covered up. They decided to investigate this phenomena.
- a** Which part of the sun's energy is believed to cause skin damage?
 - b** Write down the question that the friends are looking for an answer to.

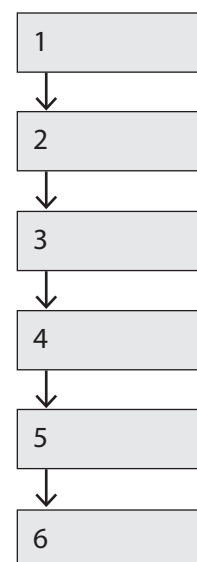
Where does a hypothesis fit into the scientific method?

Activity 2

- 2.1** Draw up a numbered flow diagram like the one shown at the right. Then write each statement into the correct place to show the accepted order of a scientific investigation.

- ▲ The hypothesis is accepted or not accepted.
- ▲ An experiment is designed to test the hypothesis.
- ▲ An interesting observation is made by a researcher.
- ▲ A hypothesis or educated guess is devised as the answer.
- ▲ The researcher asks questions in order to explain the observation.
- ▲ The experiment is conducted.

- 2.2** *In science nothing can be proved - only disproved.* Is this statement true or false?



The recipe for a hypothesis

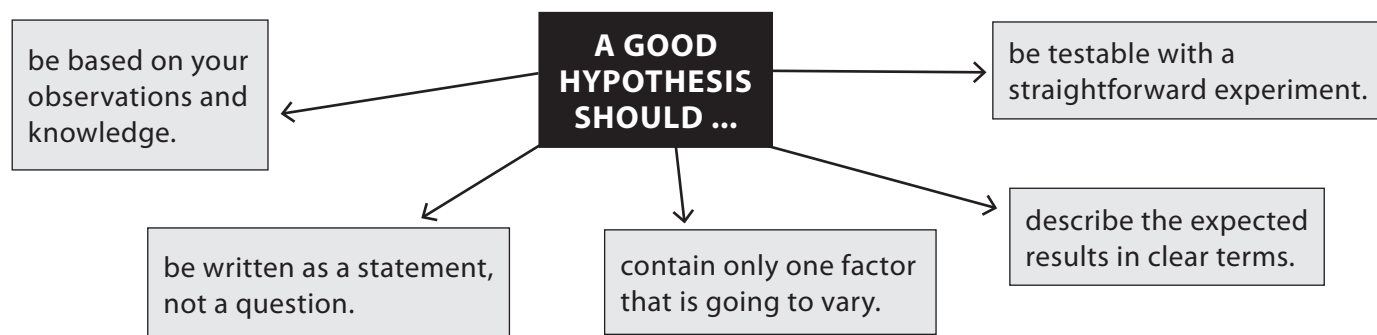
When coming up with a hypothesis, combine all the available observations with all the current knowledge about the particular phenomenon or concept. Then add a little previous experience, a pinch of intuition, some creative thinking and a flash of inspiration!

Activity 3

- 3.1** Think about the following questions and make some notes about each one.
- a** What have you observed generally about the exposure of human skin to the Sun's rays?
 - b** What do you know already about the exposure of human skin to the Sun's rays?
 - c** What experiences have you had with the exposure of your skin to the Sun's rays?
 - d** What does your own intuition tell you about the consequences of over-exposing your skin to the rays of the Sun?

Devising a hypothesis

Hypothesis formulation is one of the most creative aspects of scientific investigation. Through science investigation, we can use what we already know to write a hypothesis. The chart below points out the features of a good hypothesis.



Remember – a hypothesis or educated guess must be able to be tested to see if it always holds true.

Activity 4

4.1 Which of the following two sentences is based on 'your observations and knowledge'?

- a** The power of UV radiation can be measured using a Zeta UV Heliometer.
- b** Sunburn is caused by over-exposure to the Sun's rays.

4.2 Which of the following two sentences contains only one factor that is going to vary?

- a** Clothing of all types protects us from the Sun's rays.
- b** Fabric that does not transmit UV radiation will protect us from the Sun's rays.

4.3 Which of the following two sentences is written as a statement?

- a** Sunscreens protect us from the Sun's rays.
- b** Do sunscreens protect us from the Sun's rays?

4.4 Which of the following two sentences is testable?

- a** Skin is damaged by over-exposure to the weather.
- b** Skin is damaged by over-exposure to UV radiation.

4.5 Which of the following two sentences describes the expected results?

- a** Skin will become red when over-exposed to UV radiation.
- b** Skin does not like to be exposed to UV radiation.

4.6 Now review the question you wrote about the friends' investigation in activity 1.1 b.

- a** Using what you have learned about writing a good hypothesis, rewrite your question as a hypothesis.
- b** Briefly describe the experiment you would use to test your hypothesis.
- c** Would the experiment result in your being able to accept or reject the hypothesis, and therefore answer 'yes' or 'no' to your original question? Explain your answer.



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

What is a variable?

Conducting **science experiments** is a way of finding answers to questions. **Questions** generate hypotheses. **Hypotheses** are tested by experiments.

Science experiments must take **variables** into account. A variable is something that changes, either in an experiment or in the natural world – for example, temperature and light.

A **constant** is a factor that does not change, it always remains the same. For example, the force of gravity on Earth and the speed of light in a vacuum are constants because these values never change.

Identifying the variables

There are often several variables that affect a situation but in a scientific experiment, only one variable is allowed to change, all the other variables must be kept exactly the same. When an experiment is repeated without changing any variables, it must produce the same results each time. So identifying which variable will change and which ones will not change is very important. Variables can sometimes be a bit tricky to identify.

Chocolate science

To make chocolate, the beans from cacao trees are harvested, fermented, dried, roasted and ground. Then they are pounded into a liquid that is called chocolate liquor - but it contains no alcohol! Two special products that result from this process are:

- cocoa butter; a smooth, solid fat that is used in both food and cosmetics
- chocolate liquor; a liquid that is pure chocolate flavour – but it contains no alcohol!

The next step is to combine the chocolate liquor with other ingredients such as condensed milk and sugar. Sometimes cocoa butter is added. Cocoa butter melts at **body temperature** and gives chocolate its creamy texture. Other fats such as palm oil and vegetable oil can be added instead of cocoa butter. These fats melt at **higher temperatures** than cocoa butter does.

Activity 1

- 1.1** A group of students was curious about which chocolates have added cocoa butter and which ones have other oils added. Before they can design an experiment to test this, they must accurately identify all the variables.
 - a** List all the variables that are likely to affect their investigation.
 - b** What type of experiment could the students use to tell which chocolates have cocoa butter and which ones have different oils?
- 1.2** The students are planning another investigation with chocolate, to find out which chocolate has the best taste. They have decided that the best chocolate should melt very fast in your mouth, the flavour should be strong, a little bitter and not too sweet. They selected four different milk chocolates that were made by the same company but sold for different prices, and set up a taste test.
 - a** Which variables have the students kept constant?
 - b** Which variable are the students changing?
 - c** What other factor, which the students have not considered, may also be important?
 - d** This investigation relies on the senses, which cannot be measured. How could you introduce measurement into this investigation?
 - e** Do you think there is a link between how good a chocolate tastes and how much it costs?

Types of variables

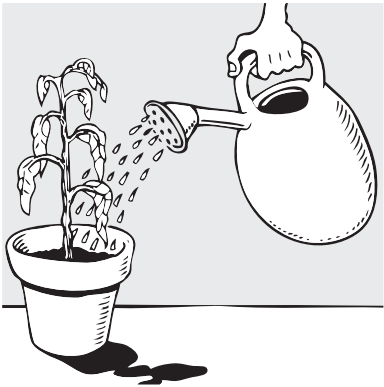
If the factors or variables that affect a situation can be separated and identified, then it is possible to design an experiment so that their individual effects can be studied.

Activity 2

2.1 Read the information about the main types of variables in the table below. Then in the last row, identify which variables in the experiment about cocoa butter in activity 1.2 go in each column.

	MANIPULATED OR INDEPENDENT VARIABLE	RESPONDING OR DEPENDENT VARIABLE	FIXED OR CONTROL VARIABLES
Variable description	The variable that you deliberately alter in some way.	The variable in which you are interested in observing a change.	Other variables that could affect the responding variable, but you keep constant.
In the investigation to find out which chocolates contain cocoa butter, the variable is ...			

2.2 Scientists are exploring how plants react to soil that is contaminated with salty ground water. They have two groups of plants that they treat in exactly the same way – same seed, same water, same amount of light, same temperature, etc. For this investigation, identify the control variables, the manipulated (independent) variable and the responding (dependent) variable.



The hypothesis – looking for cause and effect

A hypothesis proposes a possible cause and effect relationship between two variables. To correctly identify the variables, you need to look closely at the hypothesis that you wish to test. There will be at least one variable that you will change from one trial to the next. Then there are variables that you must keep the same for all the trials. You must know whether the variable you are changing is the correct one for testing the hypothesis.

Activity 3

3.1 Study the hypotheses listed in the table and fill in the variables for each one.

	HYPOTHESIS	INDEPENDENT VARIABLE	DEPENDENT VARIABLE	CONTROLLED VARIABLES
a	Salt dissolves faster in hot water than in cold water.			
b	A hard sporting ball (e.g. cricket ball) bounces higher than a soft ball (e.g. tennis ball).			
c	Drips coming out of different taps are all the same size.			
d	Babies born in winter are heavier than those born in summer.			

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.



An experiment is a practical scientific test

Once predictions – that is, **hypotheses** – have been made and the variables in a hypothesis identified, it is possible to design an experiment to study the possible influence of these variables.

Experiments are designed to test the effect of one variable, called the *independent* variable, on another variable which is called the *dependent* variable.

Checklist for experimental design

At the planning stage, consider how the hypothesis can be tested. For instance, do you have the resources, equipment and situation you need in order to physically conduct the investigation? Once this has been established then work can proceed.

Scientists start designing an investigation by writing a series of steps to be followed. This is called the **method**. The method outlines how the parts of the experiment will be conducted and in what order.

Activity 1

- 1.1** The following steps of a method are from an experiment investigating how electricity could be made by producing biogas from garbage. Steps B, C and D are out of order. What is the correct order for these three steps?

- Step A** As garbage rots, it produces biogas.
- Step B** A generator produces electricity.
- Step C** Biogas is used as a fuel to run a generator.
- Step D** Biogas is pumped into generators.
- Step E** Electricity is networked to other places.

Activity 2

- 2.1** A combination of deforestation and the burning of fossil fuels is considered to have contributed to the lower atmospheric oxygen levels found in major cities. Joseph Priestley was a biologist who carried out an experiment that showed that plants growing in sunlight could replace the oxygen that was used up by a burning candle. Choose the most logical order for the five steps in his experiment.

- Step V** After a short time the candle went out.
- Step W** Focused the Sun's rays on the candle wick – the candle burned for some time.
- Step X** A plant was pushed into the jar.
- Step Y** A glass jar was placed over the burning candle but above the water level.
- Step Z** The jar was left in the sunlight for several days.

Activity 3

- 3.1** The sequence of steps below shows the events in a biological experiment. Study the parts of the experiment and the four statements that follow on the next page.

1a. Yeast containing human antibody DNA

1b. Cell removed from a mouse embryo

2. Mouse cell now carrying human DNA

3. Cell put back into mouse embryo

4. Embryo grows inside mouse mother

5. Baby mouse carries human DNA and makes human antibodies

STATEMENTS

- Mice can make human antibodies.
- Mice do not die if given human antibodies.
- Yeast cells contain human antibody DNA.
- Human DNA can make human antibodies.

- a Which statement would be correct but not a sensible reason for doing this biological experiment?
- b Which statement is incorrect because the information was already available?
- c Which statement is obvious and not a result of experimental procedure?
- d Which statement outlines the purpose of the experiment and is based on experimental procedure?

Activity 4

Students were interested in the biodegradability of toilet paper. Research conducted by the students tells them that a biodegradable material can be broken down by natural processes and that increasing the surface area (fragmentation) of a biodegradable material increases the rate of its break down.

4.1 The following procedure or method is a series of steps followed by two students to test the hypothesis: – Cheap toilet paper breaks up in water more easily than expensive toilet paper.

What is the correct order for steps a to e below?

- a Shake both jars several times.
- b Put a sample of each in two different screw-cap jars.
- c Select samples of toilet paper, one cheap and one expensive.
- d Observe how well the paper is broken up (fragmented).
- e Add water to both jars and screw on the lids.

4.2 Identify and separate the variables involved in the students’ toilet paper experiment by completing the following table.

INDEPENDENT VARIABLE	DEPENDENT VARIABLE	CONTROL VARIABLES

- 4.3 Are all the variables in this experiment controlled, except the one to be studied?
- 4.4 Can you be reasonably sure that any differences found are due to the one variable being tested?
- 4.5 After reading the research provided above, why do you think that this hypothesis is being tested?
- 4.6 What observations are to be made during the experiment?
- 4.7 Will the observations involve counting or measuring?
- 4.8 Do you believe that the planned experiment can actually be carried out? Explain your answer.

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

Sense organs at work

People depend on all five senses to gather information about the world around them. The brain then processes this information, and tells people how to respond. Human sense organs are good, but they cannot always be trusted and they are not always reliable or accurate enough for scientific observations. Scientists use tools or equipment to enhance their observation skills.

Activity 1

- 1.1 Fill in the gaps:
A magnifying glass is a tool used by scientists to assist the sense of _____
and an amplifier helps the sense of _____.
- 1.2 We rely on our senses to detect changes in our environment. Instruments can be used to provide more precise information.
Use the following terms to complete the table by matching each sensing device with what it senses.

• set off a burglar alarm

• smoke

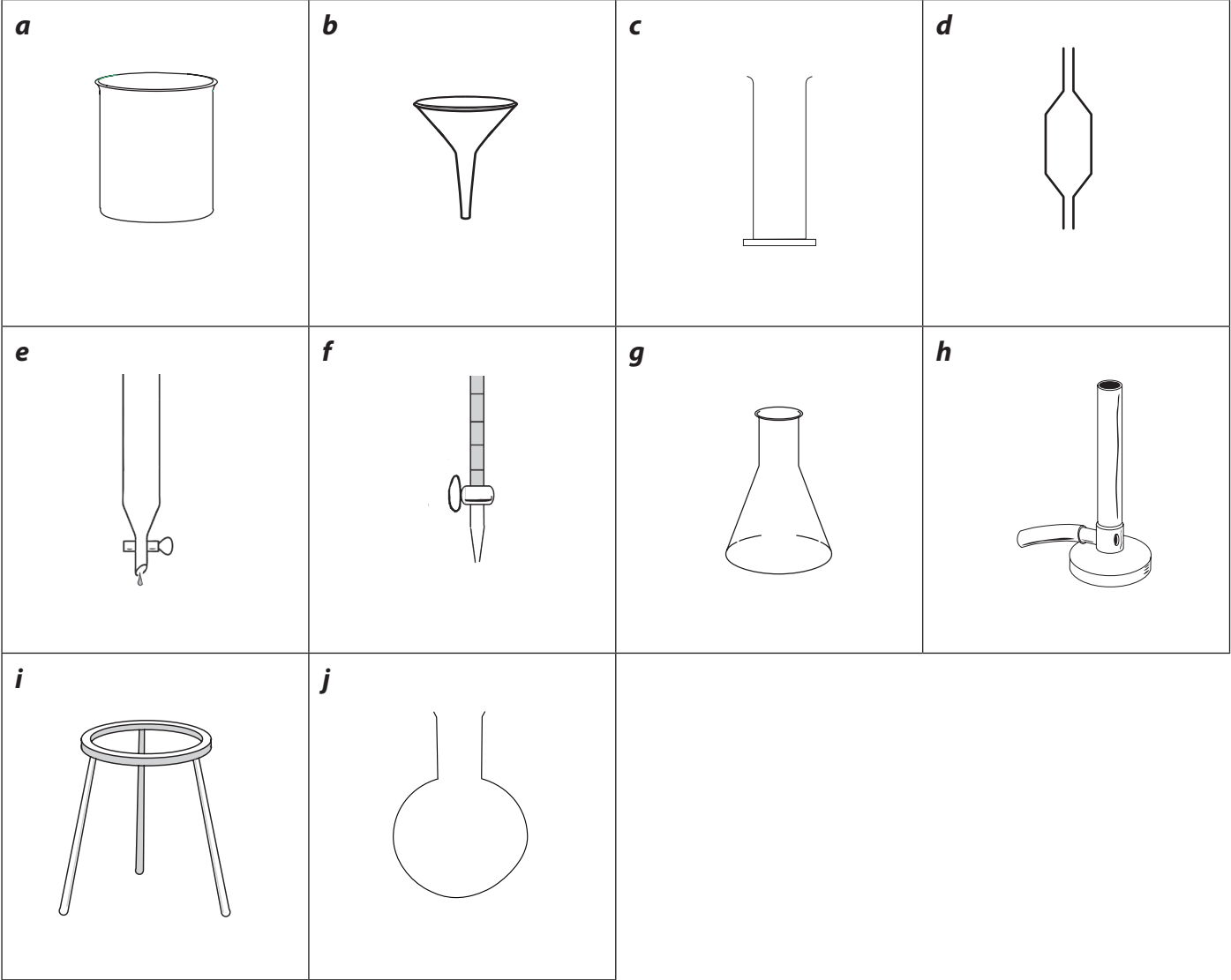
• keep air conditioners at the required temperature

• changes in temperature

• movement

• set off fire-extinguishing systems
- | WHAT IS THE TOOL? | WHAT DOES IT SENSE? | WHAT IS THE REACTION? |
|---------------------|---------------------|-----------------------|
| Photoelectric cells | | |
| Smoke detectors | | |
| Thermostats | | |
- Laboratory apparatus
- A laboratory is the place where the work of a scientist is carried out. Scientists use tools or equipment to make their work easier and their measurements more precise. Equipment used in the laboratory is called **apparatus**. In textbooks, apparatus is usually drawn in a specific way. Some work can be done with very simple apparatus, and other work requires very sophisticated and complex equipment. There are right and wrong ways to use apparatus and improper use can lead to mistakes, incorrect results and even disasters.
- Activity 2
- 2.1 From the lists below, write the correct name and use for each item of equipment shown on the next page.
Note that one use is deliberately shown twice.
- | Equipment name | Equipment use |
|------------------------|--|
| ▲ conical flask | ▲ holding liquids |
| ▲ Bunsen burner | ▲ separating solids and liquids |
| ▲ pipette | ▲ collecting and storing a gas |
| ▲ tap funnel | ▲ dispensing an accurate volume of liquid |
| ▲ burette | ▲ adding a controlled volume of a liquid |
| ▲ filter funnel | ▲ adding accurate volumes of liquid |
| ▲ gas jar | ▲ holding liquids |
| ▲ beaker | ▲ supplying heat |
| ▲ tripod | ▲ as a stand for flasks and beakers during heating |
| ▲ round-bottomed flask | ▲ heating liquids |
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Choosing instruments

Measuring makes it possible to increase the precision of observations. It also provides a way to record observations. The instruments used to measure the results of an experiment must be chosen carefully.

Activity 3

3.1 Complete the table by writing each of the following measurements beside the instrument you would choose to measure it.

Measurement: time, distance, weight, air pressure, volume.

INSTRUMENT		MEASUREMENT
a	Measuring cylinder	
b	Barometer	
c	Ruler	
d	Stop watch	
e	Electronic balance	

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

Collecting data

A collection of measurements is called **data**.

Science depends on the careful and systematic collection of measurements. These measurements may be made in a controlled setting like a laboratory or made out in the field; for example, when measuring distances in space or counting wild animal populations.

Units of measurement

Measurements are essential for comparison. In order to compare objects, scientists use standard units of measurement for all variables. For example, the standard unit for length is 'metre', for volume it is 'litre' and for weight it is 'grams' or 'kilograms'.

Activity 1

1.1 Complete the table by filling in the correct information at each '?'.

QUANTITY	UNIT OF MEASUREMENT	ABBREVIATION	RELATIONSHIP BETWEEN UNITS
Mass	kilogram gram	kg g	1000 g = 1 kg
Area	square metre ?	? ?	1 m ² = 10 000 cm ²
Energy	joule ?	J kJ	?
Temperature	degrees Celsius	?	(n/a)
Time	? minute	s min	?

1.2 It is important to choose units of measurement that are suitable for the quantities involved in an experiment. For example, if an experiment involves measuring small quantities of energy then joules should be used. For large quantities of energy, kilojoules should be used. Which units would you use to measure these quantities?

- a Diameter of mould colonies
- b Weight of a person
- c Area of a country
- d Volume of water in a pool
- e Distance in space



SI units

SI units are those specified in the International System of Units (abbreviated SI from the French, *Le Système International d'Unités*). These units have been chosen by an international committee for measuring specific quantities. For example, the SI unit for force is the Newton (N) and for distance it is the metre (m).

It is important to remember that if measurements gained from an experiment are to be used in an equation, they must first be converted into the appropriate SI units.

Activity 2

2.1 The International System of Units has been in place since 1960. Why was it important to develop a worldwide system of measurement for use in science and commerce?

2.2 The volume of regular shaped solids can be found using a ruler and a mathematical formula.

- a** What is the volume of a rectangular block which is 5 cm long, 4 cm wide and 3 cm deep?
- b** What formula did you use?

2.3 Finding the density of an object also involves using an equation: $\text{density} = \frac{\text{mass}}{\text{volume}}$

- a** What is the density of a rock with mass 25 grams and volume 10 cubic centimetres?
- b** In what unit is your result measured?
- c** What would your results be in kilograms per cubic metres?

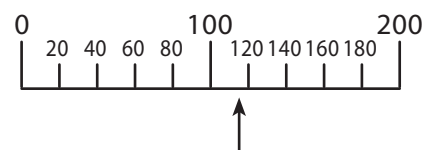
Measuring error

No matter how careful scientists are, measurement errors can occur in scientific experiments. Measuring instruments are only accurate to a certain degree, and different people may read the scale on a particular instrument slightly differently. Scientists overcome these kinds of measurement uncertainties by repeating an experiment several times, making repeated measurements of the same quantity and by using averaging techniques.

To work out the uncertainty of a measurement, note the smallest division in the scale and divide it by two. For example, look at a standard 30 cm ruler. The smallest division is 1 mm. Divide this by 2 and you get 0.5 mm. This is the uncertainty and it is written as $\pm 0.5 \text{ mm}$. Therefore, a measurement on your ruler of 7 mm written expressing this uncertainty would be $7 \text{ mm} \pm 0.5 \text{ mm}$. In real terms, this means that the measurement is between 6.5 mm and 7.5 mm.

Activity 3

3.1 The scale at the right displays a measure of pressure (kPa). Write the measurement shown by the arrow, and include the uncertainty.



3.2 Sometimes the uncertainty is not limited to a single value.

Suppose you are measuring your pulse rate with a stop watch. The error may be in your reaction time as you press start and stop on the stopwatch, rather than in the precision of the stop watch. To find an estimate of the uncertainty in your reaction time, follow the steps below.

- a** Estimate your reaction time by starting and stopping the stopwatch as quickly as you can. Sample data, measured in seconds, is shown below.

0.21 s	0.19 s	0.20 s	0.25 s	0.23 s	0.17 s	0.22 s	0.20 s	0.22 s	0.21 s
--------	--------	--------	--------	--------	--------	--------	--------	--------	--------

- b** Add all the measurements to get the total value. What is the answer?
- c** Divide this total value by the number of measurements. This is the average value.
- d** Look at the set of measurements and note the highest and the lowest numbers. What are they?
- e** Subtract the average value from the largest value. What is it?
- f** Subtract the smallest value from the average value. What is it?
- g** Add together the values obtained in (e) and (f) and divide by 2. What is your answer?
- h** Record your measurement as well as its uncertainty: average value $\pm$ uncertainty.

3.3 A microbiologist is measuring the number of bacteria present in a milk sample. Work out the average value and the uncertainty in this group of measurements.

Bacterial count	20	13	14	19	18	17
-----------------	----	----	----	----	----	----

3.4 In your own words, explain why considering measurement uncertainty is essential in science.



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

Put your boots on!

Most experiments are carried out in a laboratory, but some need to be conducted outside 'in the field'. The advantage of field experiments is that the observations are conducted in a natural setting rather than in a contrived laboratory environment. However, in field experiments, conditions cannot be controlled with the same precision and certainty as they can in a laboratory.

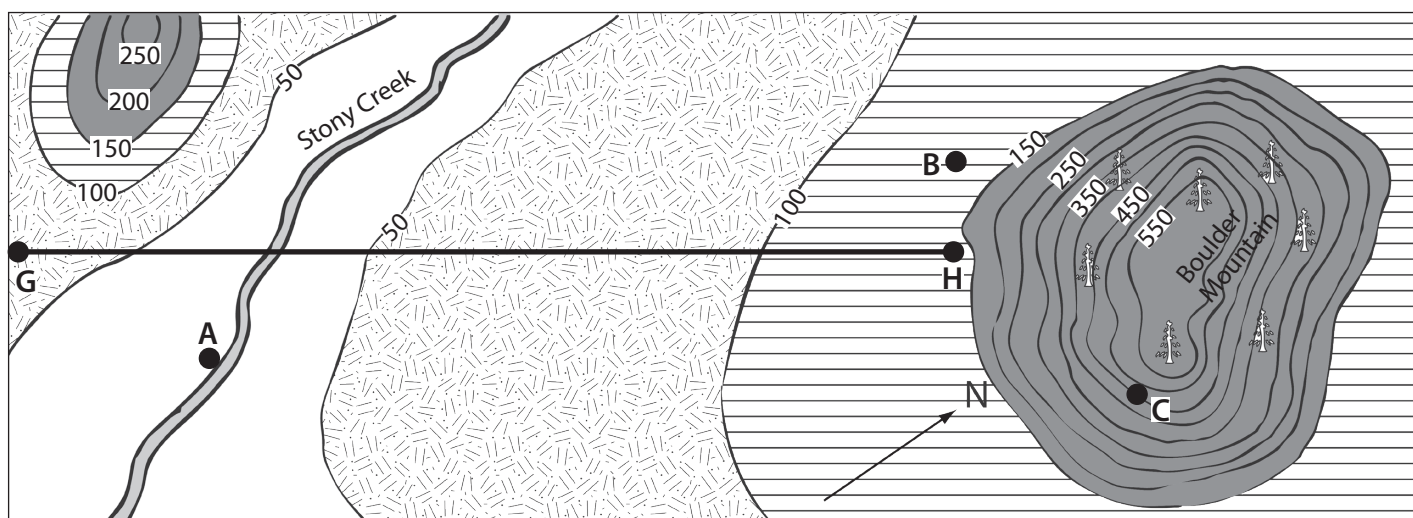
Activity 1

1.1 *Lantana camara*, commonly referred to as lantana, was introduced to Australia as a garden plant in 1841. Since then, it has become a widespread bushland weed. Lantana prefers good soil moisture, warm conditions and protection from strong winds. Plants usually have natural enemies, such as insects and diseases. There are several natural insect species that affect lantana. Scientists are investigating the effects of a locally found predator insect on bushland lantana.

They hope to determine the abundance of the insects on the lantana plants and the amount of damage inflicted on the bushes. The insects do the most damage to unhealthy plants that are growing where it is warm and out of the wind.

The scientists chose three different sites within a small range. These sites are marked on the map below.

- ▲ Site A – Lantana growing in the bed of Stony Creek which had dried up due to drought
- ▲ Site B – A band of bushes at the base of Boulder Mountain
- ▲ Site C – A group of plants at an elevation of 450 metres above sea level on Boulder Mountain



To estimate walking time allow 1 hour for:

- ▲ each 5 km of easy going
- ▲ each 3 km of gently sloping ground
- ▲ each 1.5 km of thick bush
- ▲ each 0.5 km of mountain climbing.

KEY:

- | | |
|-------------------------------|--|
| Flat ground with weed patches | Gently sloping ground with sparse shrubs |
| Gentle hills with thick bush | Steep mountain with sparse tree cover |

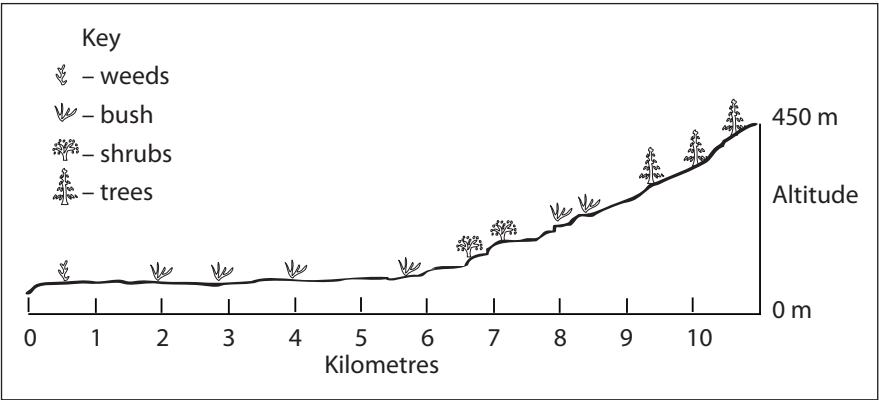
- a** The scientists had base camps marked at G and H on the map.
How long would it take them to walk from Camp G to Camp H?
- b** How long would it take them to get to Site C from Camp H and back again?

1.2 The scientists were working in collaboration with a group of ecologists, and agreed to record the types of native vegetation growing at different altitudes. Record this data in the table for the ecologists.

ALTITUDE (m)	VEGETATION ON SOUTH-FACING SLOPES
550–150	
150–100	
100–50	
50–0	

1.3 A transect is a straight line through an area. It shows how plants change as the geography of an area changes. Study the ‘elevation view’ of a transect through the study area shown at the right.

Between which two points on the map was the information recorded – A and H, H and C, G and B or A and C?



1.4 Three lantana plants were studied at each site – A, B and C. Within each plant, three sampling units (a branch with about 20 leaves and of a standard length) were selected for observation. The data collected for each site was averaged and is presented in the table.

DATA COLLECTED	SITE A	SITE B	SITE C
Leaf area (cm ²)	4	6	4.5
Buds	3	10	8
Flowers	3	11	8
Fruiting heads	2	10	7
Number of insects	50	15	25
Leaf damage (%)	65	10	30
Evidence of insects (eggs)	yes	no	yes

- a On which site was overall insect attack the least?
- b Why did the scientists record so much information about the actual plants?

1.5 Consider the information below about the three sites. Then suggest why the plants at Site A were so badly infested.

SITE A	SITE B	SITE C
Soil moisture is low, ground temperatures are high, winds absent.	Soil moisture is good, ground temperatures are medium and strong winds are rare.	Soil moisture is average, ground temperatures are medium to low and winds are medium to high.

1.6 Scientists are searching the country in which lantana originated for its natural insect predator. They are hoping that by importing it, testing it and releasing it, the natural predator will do a better job of managing lantana in bushland areas than the local predator. This weed management method is called biological control.

Their research indicates that the imported predator prefers altitudes between 100 and 400 metres above sea level, where there is protection from winds and hot dry conditions. Where on the transect diagram do you think this predator will have the greatest weed management effect?

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

Data-handling skills

Scientists communicate in many ways. One way to present scientific information is in non-textual, two-dimensional formats such as sketches, photographs, diagrams, maps, plans, charts and graphs. These formats offer three major advantages in comparison to written or verbally presented information. They are concise and can set a scene immediately, they are memorable and they make relationships within the information readily apparent.

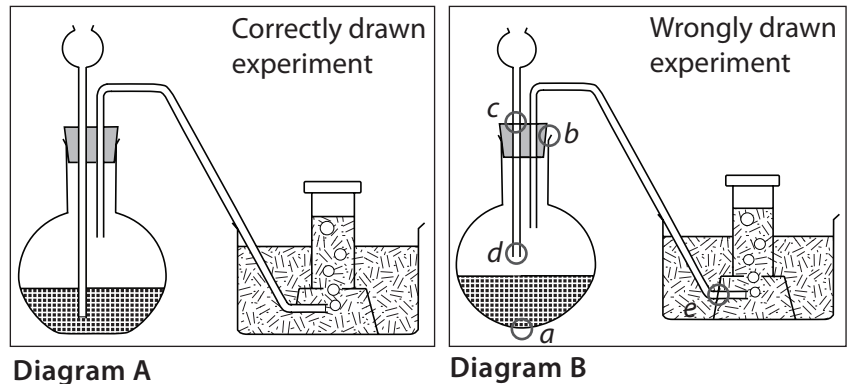
Diagrams

When reporting on an investigation, a diagram is often required – for example, laboratory apparatus, the operation of a piece of equipment or the drawing of an organism. Diagrams must be drawn accurately and be clearly labelled.

Activity 1

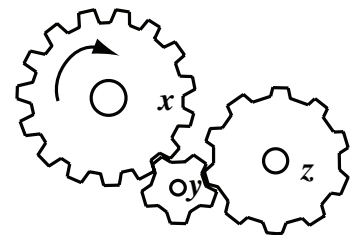
- 1.1** Study the two diagrams of the hydrogen experiment. Diagram A is correctly drawn and diagram B is incorrectly drawn.

Compare the two diagrams and list the five mistakes in diagram B. These are indicated by the circles at a to e.



- 1.2** Arrows can be used on diagrams for labelling, to show movement, to indicate direction and to sequence points. The diagram below right shows three wheels which are free to rotate. Wheel X has 15 teeth, wheel Y has 5 teeth and wheel Z has 10 teeth.

- a** As the arrow indicates, wheel X is rotating in a clockwise direction. Add arrows to show which way wheels Y and Z are rotating.
- b** Which wheel will rotate faster – Y or Z?
- c** What is the purpose of the arrows in this diagram?
- d** Complete the statement so that it is correct:



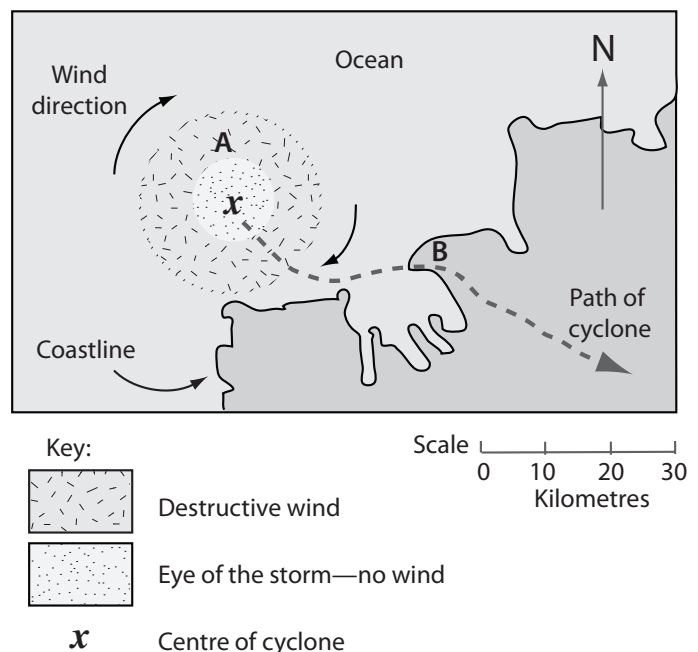
When wheel X turns, wheel Y will be going _____ times as fast as wheel X, _____ will be going _____ as fast as wheel Y and _____ will be going half as fast as wheel X.

Maps and charts

Maps are easy to remember, but keys and symbols must be clear and understood before maps can make sense.

Activity 2

- 2.1** The map shows the path of a cyclone as it passes over the coastline of a land mass in the tropics.
- a** Is the cyclone moving anticlockwise or clockwise?
 - b** In which direction is the cyclone moving?
 - c** What is the diameter of the cyclone?
 - d** Describe the type of wind blowing in section A.



- e What symbol marks the centre of the cyclone?
- f What symbol represents the path of the cyclone?

Using and interpreting lines on diagrams

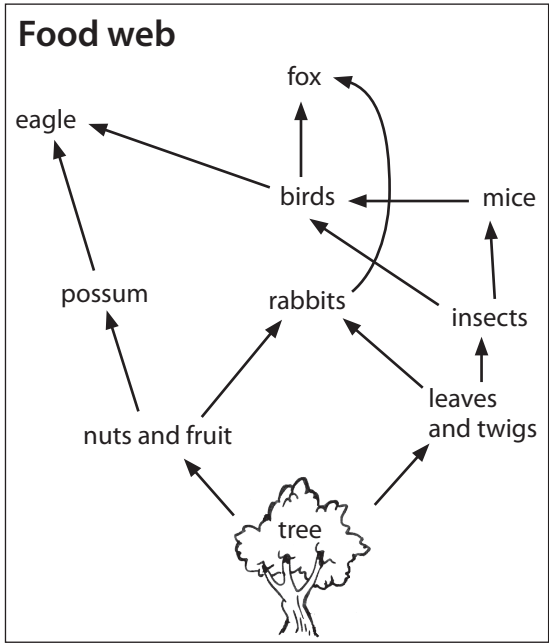
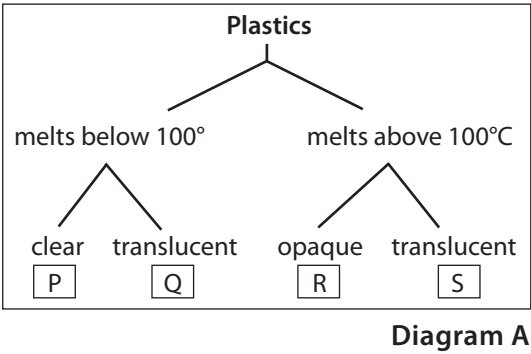
Sometimes lines and arrows on diagrams are not supposed to be interpreted literally but rather as a 'commentary' or as explanatory 'notes' for the diagram. For example, the lines used in the diagram at the right 'explain' the bat's use of sonar. These lines could also be used to represent heat, sound or the motion of a moving object.



Activity 3

3.1 The lines in the diagrams at the right display the relationships between items.

- a What do the lines represent in diagram A?
- b What do the arrows represent in diagram B?
- c In diagram B, what does the direction in which the arrows are pointing indicate?

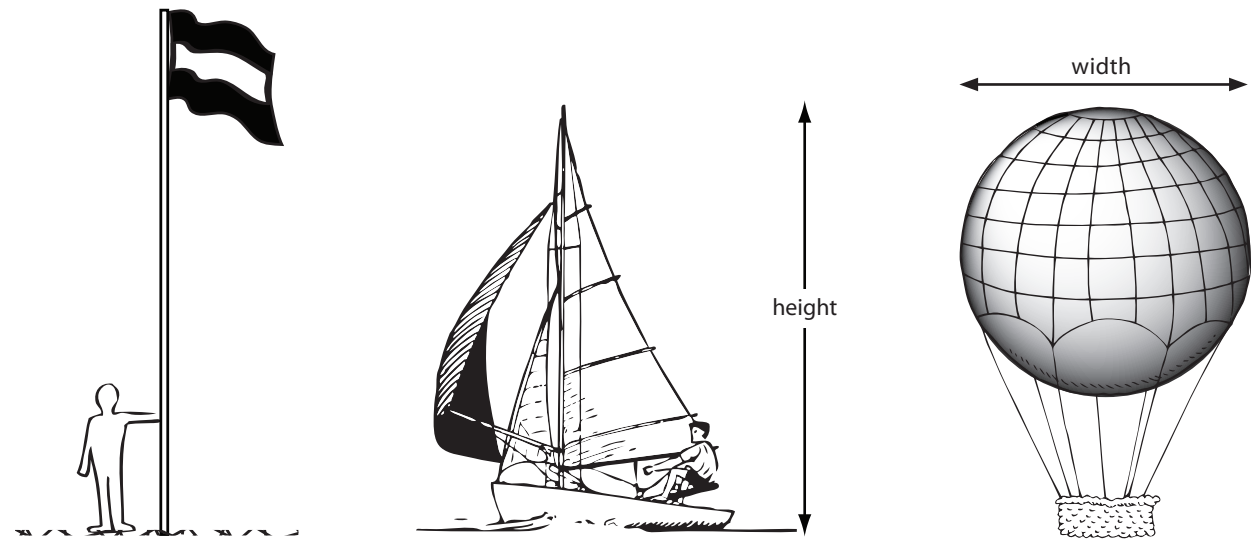


Scale drawings

Scale drawings and diagrams are a practical application of ratios.

Activity 4

4.1 Study the diagrams below, which have been drawn to the same scale.



- a The person in the first diagram is two metres tall. What is the height of the flagpole?
- b What is the scale ratio in the first diagram?
- c Using this scale, what is the real height of the sail boat and the real width of the hot air balloon?
- d If a scale ratio of 1 cm representing 5 m had been used, how tall would the sail boat be?

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

Displaying quantitative information

A table is used to display data in an organised way that is easy to understand. It is this data that is used to construct graphs. Graphs allow data to be compared visually. They can be used effectively to communicate relationships. Graphs are not superfluous or an alternative to tables as a way of representing numerical data. Rather, graphs are an essential part of understanding, interpreting and communicating the results of a scientific investigation.

Discrete and continuous data

The observations and measurements from science experiments are recorded as discrete or continuous data.

- Discrete data** is in categories. For example, gender, species of animal, brand of car or colour of lipstick. The values are distinct and separate, and are counted in whole numbers. For example, you cannot have 5.2 birds or 3.1 brands of car.
- Continuous data** is associated with measurements that involve a standard scale with equal intervals. For example, height of trees in metres (7.3 m), the amount of chocolate in grams (95.5 g) or the length of time in minutes (2.5 min).

Activity 1

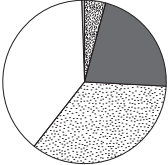
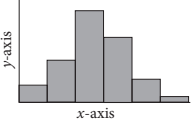
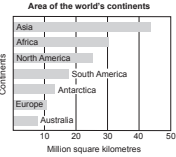
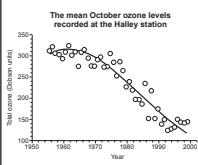
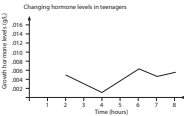
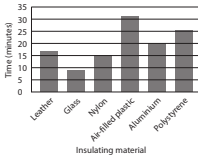
1.1 Identify each variable in the table below as discrete data or continuous data.

VARIABLE TO BE MEASURED	DATA – DISCRETE OR CONTINUOUS?
a Brand of sports shoe	
b Number of whole pizzas	
c ‘Mack’ trucks on a highway	
d Temperature of the surf	
e Colours of nail polish	

1.2 There are many different ways to graphically show the relationship between variables. An important aspect of graphing skills is the ability to select the graph format that best suits the information to be presented.

Complete the table below so that the main graphical types are summarised correctly. Write the correct graph name from the list below, decide whether the data shown is discrete or continuous, then write the matching letter (A to F) from the graph descriptions on the next page.

Graph names: line graph, pie chart, bar graph, column graph, scattergram, histogram

Graph format						
Graph name						
Discrete or continuous data?						
Description (letter)						

GRAPHS DESCRIPTIONS

- A** Makes a visual comparison between several sets of results. These graphs are used when one value is not numerical.
- B** The horizontal axis is marked in equal intervals, columns are drawn for the relevant frequency and adjacent columns have a common edge.
- C** Vertical columns of equal width are drawn on the x-axis, to the appropriate height of the vertical scale.
- D** A circle divided into sectors where the angles at the centre are in the same proportion as the component parts. These graphs are used for displaying the proportions of a single entity.
- E** For displaying the distribution of two variables, which may not have a dependent relationship.
- F** For displaying the relationship between two variables for which the obtained data are samples of a continuum.

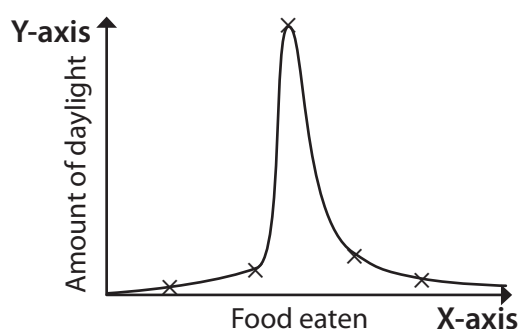
RULES FOR DRAWING GRAPHS

- Rule 1:** The graph must have a title.
- Rule 2:** The x-axis should display the independent variable (experimenter controls the cause) and the y-axis should display the dependent variable (experimenter measures the effect).
- Rule 3:** The axes should be similar in length and be clearly labelled with the variable they represent, together with the units of measurement.
- Rule 4:** The scale on both axes should be in equal, sequential intervals and start from zero (the origin) where possible.
- Rule 5:** If there is a large gap between zero and the first reading, the axis can be broken immediately after the zero and before the first data.
- Rule 6:** Plot each point accurately with 'x' or 'o' and join the points with a line to demonstrate the trend. Unless it can be reasonably assumed to be correct, do not extend the line to the origin if the data does not start at zero.

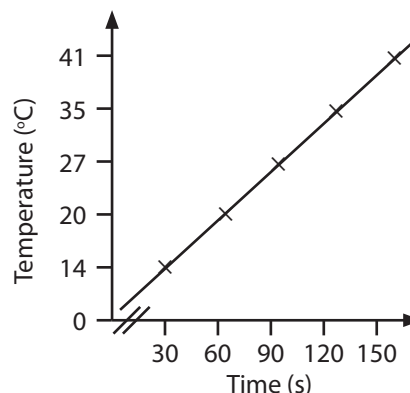
Activity 2

2.1 Read the rules for drawing graphs above, then look at the sample graphs below. For each graph, list the rule numbers that have been broken.

- a** Graph 'A' shows the amount of food eaten by a wallaby between midnight and midday. Wallabies like to eat just before dawn.



- b** Graph 'B' shows the day's temperature for the 2.5 hour period following dawn.



2.2 Which type of graph would be the most suitable for presenting the following data? Refer to the table in activity 1.2 to help you.

- a** Percentage of each bird species that makes up the bird population of a national park.
- b** Body weight of calves as they grow over twelve months.
- c** Average time that runners took to run four separate races.
- d** Heights of students in a class.
- e** The number of peas found in various pea pods.

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.



What do the results show?

Scientists constantly seek evidence to support and substantiate their ideas and predictions. In science, people look for connections in nature and patterns in data. This increases our understanding of the natural world, and our ability to manipulate it.

Looking for connections and seeking patterns

Over time, the ability of the human mind to seek patterns has developed and been refined. For example, humans had to know where food grew in different seasons, when and where water was available, and signs that indicated danger – these skills were all crucial to people's survival.

Modern humans have an overwhelming urge to seek patterns, and we are quite good at it! Scientists apply this skill to recognising cause and effect relationships and pin-pointing nature's regularities.

Activity 1

Identifying patterns and working with patterns involves looking for relationships between things.

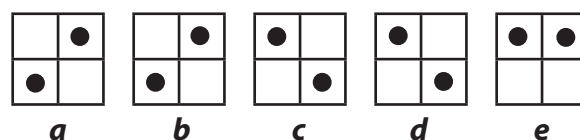
1.1 Which is the odd one out in each of these lists?

a galah, seagull, cockatoo, budgie

b wine, water, milk, bread, tea

c New York, Washington, San Francisco, London, Seattle

1.2 Which of the boxes at the right does not fit the sequence?



1.3 Examine this series of letters: X Y Y X Z X Y Z X Y Y X Z X Y.

What should the next three letters be?

1.4 Scientists often group things according to similarities. Look around you. Group the things you can see according to what they are made of: *wood, paper, plastic, metal, organic*.

Checklist for evaluating evidence

When scientists evaluate evidence, they search their data for valid support for their predictions and explanations. They may need to calculate answers using mathematical formula and display the results in simple diagrams, sketches, charts or graphs so that they can see any relationships more clearly.

Activity 2

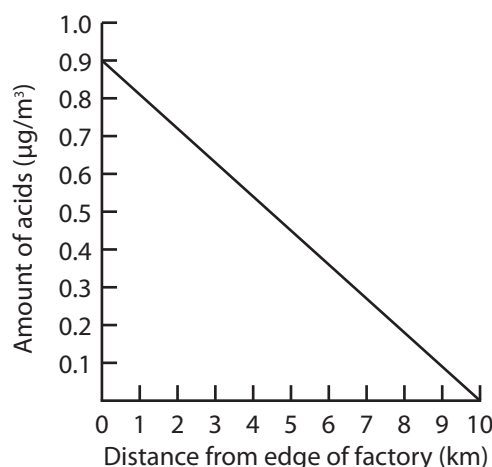
There is a fertiliser factory operating at the mouth of a river near a large city. Fertiliser factory emissions contain weak acids which are known to damage metal.

Proposals have been lodged for three developments to be constructed nearby:

- ▲ A large car yard – major metal will be iron
- ▲ A residential area – major metal will be aluminium
- ▲ A boat marina – major metal will be stainless steel

2.1 A scientist was asked to investigate whether the fertiliser factory would cause damage to the surrounding area. She obtained three plates – one iron, one aluminum and one stainless steel – and positioned them five kilometres from the factory. The plates each measured 1 m². Over a period of two months, she measured the area of damage to the plates in cm². The scientist's results are shown in the graph at the right and in the table on the next page.

Distance from the fertiliser factory and amount of weak acid in the air



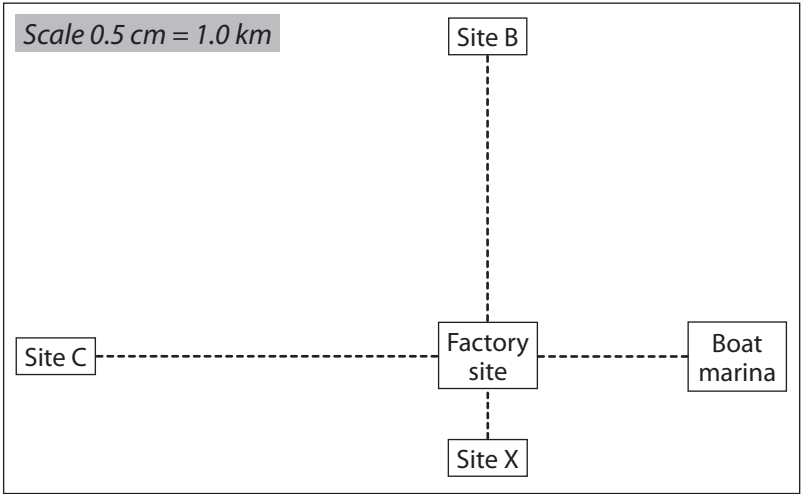
a When the amount of weak acid is 0.3 ug/m³, which metal shows the most damage?

TABLE OF AMOUNT OF WEAK ACID AND DAMAGE TO METALS

Amount of weak acid in the air (ug/m ³)	Car yard	Residential area	Boat marina
	Iron damage (cm ²)	Aluminum damage (cm ²)	Stainless steel damage (cm ²)
0	3	4	22
0.1	5	14	23
0.3	10	25	21
0.5	20	34	20
0.7	42	42	22
0.9	60	48	21

- b From the diagram at the right, what is the shortest distance in kilometres between the factory and the boat marina?
- c If there are samples of iron, aluminium and stainless steel at the boat marina, which type of metal will show the most damage?
- d Based on expected metal damage, which development is best suited to:
- ▲ site B?

▲ site X?



- 2.2 Draw graphs to show the damage for each type of metal, according to the amount of acid in the air.
- Use the same set of axes for all three graphs, with metal damage on the vertical axis and amount of acid in the air on the horizontal axis.
- a Study your graph. Explain the damage occurring to stainless steel.
- b What does the graph tell you about the rate of damage to aluminum due to acid levels?
- c The iron graph shows a changing slope, which suggests a changing rate of increasing damage. Where on the iron graph is the rate of damage the highest?
- d What would be the level of damage per square metre to the iron roof of the fertiliser factory?

Activity 3

- 3.1 Use your analysis of the investigation in activity 2 to write your assessment of the investigation. Use the evaluation checklist criteria listed below.
- a Make a relevant comment about the suitability of the basic method used.
- b Comment on the accuracy and reliability of the results.
- c Suggest changes to improve the reliability of the evidence.
- d Find trends and patterns in the results and explain any results that do not fit the trend.
- e Explain simply what was discovered – the conclusion.
- f Is the evidence good enough to support a firm conclusion?
- g Propose improvements or further work, that would strengthen the conclusion.

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

Communication counts

Communication is important in science. One reason for this is that the more complex our world becomes, the more we need to understand in order to make responsible decisions. Scientists always tell other scientists about their work. The way scientists share their knowledge and allow others to build on it and develop it further is a behaviour that is taken for granted, but it is very special and important.

How scientists communicate – science reports

Scientists tell people about their work. It is important for other investigators to independently test the same hypothesis, and replicate the results in order to verify them. To do this, the investigators have to follow the original procedure exactly and have access to the original data.

Scientific reports are published in scientific journals. The organisation that publishes the reports generally sets requirements for the format and style of the report. However, a scientific report about practical research usually follows a standard format.

Activity 1

- 1.1 Why is it important for scientific reports to have a standard format?
- 1.2 Writing a scientific report can be compared to writing a story.
Read the ‘story’ below and fill in the term that explains each section.
Terms: findings, aim, contribution to knowledge, method, literature review

Once upon a time researchers believed ... (); but then I thought that maybe ... (); so what I did was ... (); and I have discovered that ... (); which changed the way we think about ... ().

- 1.3 Draw lines to match each section of the report with the information that should be included in it.

SECTION OF REPORT	INFORMATION INCLUDED
a Title	▲ What do I hope to discover?
b Introduction	▲ What does it all mean?
c Review of literature	▲ What did I find out? What are the possible applications and recommendations? Do the results support the hypothesis or not? What next?
d Hypothesis	▲ How do I describe the content of this report in the smallest number of words?
e Method	▲ What do we know? What don't we know?
f Results	▲ What have I found out?
g Discussion	▲ Why am I doing it?
h Conclusion	▲ How am I going to discover it and what equipment do I need?

The ‘Rules’

There are general ‘rules’ for writing a scientific report, which you will explore in activity 2.

Activity 2

2.1 Read the ‘Scientific’ Report below. Then consider and complete the following tasks **a** to **e** to help you analyse it.

‘SCIENTIFIC’ REPORT

My friend and I want to find out which plants still grow in a place where the pH of the soil has been increased by alkaline run-off from a building site. We gathered together things like a GPS, a pH meter and a plant identification book. We took the long way around to the site as it was a lovely walk. When we got there, I decided to record the plants that we found in each quadrat and my friend recorded the pH values in the quadrats. We also noted the exact location of each quadrat using the GPS and the tape measure. In our opinion, the high pH levels only stopped the growth of some of the plants, not all of them.

- a** All scientific reports are written in the third person. This means that you should not write a report in terms of ‘I’, ‘you’ or ‘we’. For example, write ‘the researcher’ instead of ‘I’ and ‘the research team’ instead of ‘we’ or ‘us’. Highlight the words you need to change in the report above so that it is in the third person.
- b** Avoid terms such as ‘in my opinion’. Instead, simply state what was found out. Circle the sections in the report that you need to alter to meet this criteria.
- c** Be precise in your use of language – use short, active sentences. Underline the sections of the report that you need to change to meet this criteria.
- d** If you use scientific terms or acronyms that the reader may not know, you should explain them either in the body of the report or in a glossary. Tick at least two terms in the report which could be clarified.
- e** Use tabulated lists rather than long sentences with lists contained within them. Tick the example below that demonstrates the accepted format.

EXAMPLE A

- ▲ Equipment used included a GPS, a 50 m tape, quadrat markers, a plant identification guide and pH testing kits.

EXAMPLE B

- Equipment
- ▲ GPS
 - ▲ 50 m tape
 - ▲ quadrat markers
 - ▲ plant identification guide
 - ▲ pH testing kits.

2.2 Now use what you discovered above to write an improved version of the Scientific Report.



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

Safety awareness

Science laboratories are potentially dangerous places because of the nature of the work that occurs inside them. When accidents do happen, it is usually because people have not used their common sense and followed the rules and observed the procedures designed to reduce risk.

Dress for the occasion!

Activity 1

1.1 Personal, protective, safety clothing and equipment is provided in most laboratories. Write the personal protective equipment (PPE) from the list in the correct row of the following table.

PPE: safety glasses, laboratory coats, rubber gloves, safety helmets and visors, face masks and respirators, splash goggles

PPE	PROTECTION FROM
a	Corrosive chemicals coming in contact with skin
b	Small but not serious liquid splashes
c	Breathing in airborne materials and vapours
d	General exposure to dangerous materials
e	General clothing protection from spills, splashes and marks
f	Large liquid splashes and serious projectiles

1.2 Tick the pieces of PPE in the table above that you use or can see in your school laboratory.

Do not taste, touch or smell chemicals – ever!

Chemicals surround us throughout our lives. Some are harmless but others are very hazardous. Chemicals can enter our bodies in four ways – inhalation, absorption, ingestion and injection.

Activity 2

- 2.1 List at least two tools that are provided in your school laboratory to help avoid chemical contact with your skin.
- 2.2 What is one piece of equipment that is found in most laboratories and is designed to stop chemical vapours entering your nose and eyes?
- 2.3 Name two pieces of equipment that are provided in most laboratories for use in situations where chemicals do come in contact with the skin or eyes.

Avoid fire and explosions!

Fire is usually caused by the incorrect use and handling of chemicals, electrical equipment or gases. The three parts of a fire 'triangle' are air (oxygen), fuel and heat. Remove any one of these elements and the fire will go out. For example, cool with water (remove heat), starve the fire of things to burn (remove fuel) or smother or dilute the air supply (remove oxygen).

Activity 3

3.1 There are several devices found in a school laboratory for extinguishing a fire, should it occur. Complete the table on the next page by writing the correct use from the following list and then identifying which part of the fire triangle is removed.

Uses: For larger fires where the fuel is known; fires on people; small fires

TYPE OF EXTINGUISHER	USE	PART OF THE FIRE TRIANGLE REMOVED
Sand bucket		
Fire blanket		
Fire extinguisher		

3.2 Study your school laboratory.

- a Locate the fire bucket, the fire extinguishers, the fire blanket and the fire alarm.
- b Assess the way these items are stored. For example, are they easily accessible? Are they clearly identified?

If you are unsure, assume it is dangerous!

Some biological materials found in science laboratories can cause infections, disease, tissue damage and some produce toxins.

Activity 4

4.1 Draw lines to identify the possible dangers in each situation.

SITUATIONS

- a Opening a petri dish in which cultures of fungi and bacteria, obtained from a toilet seat, are growing.
- b Touching your eyes after dissecting plant material.
- c Forgetting to wash your hands after dissecting the gut of an animal.
- d Touching animals that are often housed in school laboratories, such as toads, fish and reptiles.

DANGERS

- Coming into contact with animal-produced toxins
- Breathing in fungal spores or bacterial cells
- Poisonous plants
- Infections with parasites

Avoid the pointy end!

The term 'sharps' means anything that is sharp, such as syringes with needles, scalpels and broken glass - they can cut you and they may be contaminated. Dispose of sharps in the container supplied for the purpose and put broken glass into the 'glass' bin.

Activity 5

5.1 Locate the 'glass bin' and the 'sharps container' in your school laboratory.

Disposal of wastes

Activity 6

6.1 Complete the text below by filling in the correct vowels.

Correct handling of laboratory waste

- a P_t r_bb_sh _n th_ c_rr_ct b_n.
- b Fl_sh _nly _ll_w_d ch_m_c_ls d_wn th_ s_nks, w_th pl_nty _f w_t_r.
- c D_ n_t _se c_nt__n_rs pr_v_d_d f_r th_r ch_m_c_ls.
- d _s_ j_st _n__gh ch_m_c_l f_r th_ pr_c_d_r_s_ th_t th_r_ _s n_th_ng l_ft _v_r f_r d_sp_s_l.

Activity 7

7.1 Draw and annotate a map of your school laboratory, identifying the possible hazards, the safety signs and the safety equipment.



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

Models make ideas more real

A model is a simplified imitation of an object, situation or idea that helps us to understand it better. A model may be a device, a plan, a drawing, an equation, a computer program or even just a mental image. Whether models are physical, mathematical or conceptual, their value is in the way they show us how things either work or might work.

Making difficult scientific concepts clearer

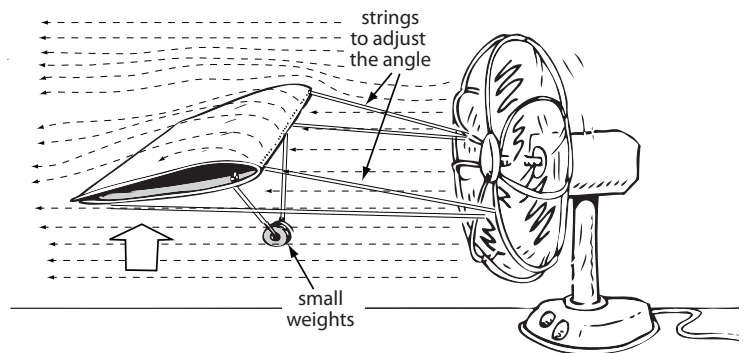
A physical model is an actual device that is easier to work with than the thing it represents. For example, it may be smaller in size or less expensive than the real thing – like a model train. Experiments using physical models allow the variables to be carefully controlled, in the hope that the results will be similar to experimenting with the real thing.

Activity 1

- 1.1** Scale models of aeroplanes are tested in wind tunnels to investigate the effects of air movement over different wing shapes. Read the information and study the model below of an aeroplane wing or aerofoil in a 'wind tunnel' provided by a fan. The model shows why wings are shaped the way they are.

The velocity of air molecules travelling up and over the wing effectively creates a reduced pressure on the upper surface of the wing. The speed of the air moving over the wing is greater, which reduces pressure and causes the plane to rise into the air (lift).

Ailerons or wing flaps on the back of a wing can be raised or lowered to cause the plane to decrease speed (drag).



Model of wind tunnel and aeroplane wing or aerofoil

- a** Why have strings been added to the angle of the wing?
 - b** What information can we investigate by adding the weights to the model wing?
 - c** How would a flat piece of card behave in comparison to the aerofoil?
 - d** If 1 cm lengths of cotton were stuck all over both the flat card and the aerofoil, how would this help to explain why the aerofoil lifts better than the flat card?
 - e** If the back edge of the wing was folded upwards, what would you expect to observe?
- 1.2** Wind tunnels are used extensively to test wind drag when designing new products. Real wind tunnels can be large enough to fit scale models. Racing skiers actually stand inside the tunnels to test equipment and to find the best way to stand in different winds.
- a** List two other sports that are affected by drag.
 - b** List the equipment used in these sports that could be tested in a wind tunnel to improve performance.
- 1.3** In testing new medical treatments, what is the advantage of using laboratory animals or cultures in test tubes to model human biological processes?
- 1.4** A model can be scaled down in time as well as in size and materials. Why are genetic experiments carried out on organisms such as bacteria, flies and mice?
- 1.5** Models do have limitations. The information gained from them must not be extended too far. Think of one limitation for each of the models described in questions 1.3 and 1.4.

Conceptual models

Conceptual models are 'mental pictures' that help us to interpret phenomena. They can be expressed in mathematical terms, as physical models and even as computer simulations. Relating something unfamiliar to something more familiar also allows us to communicate these mental pictures or ideas. Analogies and metaphors are also used for comparison; for example, a computer 'virus', a description about a 'blanket' of cloud over the sky and the way cars were once referred to as 'horseless carriages'.

Activity 2

- 2.1** Chemistry involves studying tiny but complicated particles called atoms. To describe them, chemists use simplified diagrams like diagrams 1 and 2 below. No one thinks that atoms actually look like this, but the diagrams show the parts that the chemists are interested in.

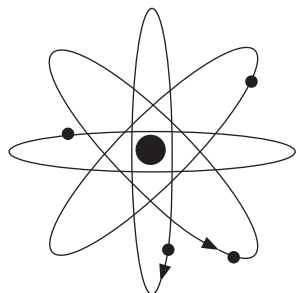


Diagram 1

The first shell outside the nucleus holds up to two electrons.

The second shell holds up to eight electrons.

The third shell holds up to eight electrons.

When a shell is full, a new one is started.

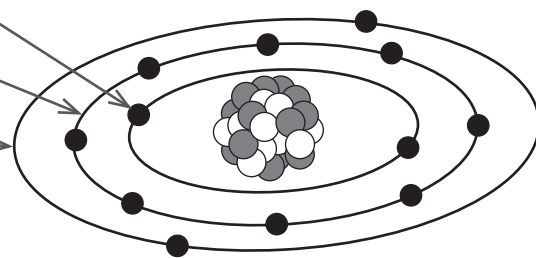


Diagram 2

- a** Name the sub-atomic parts found in the nucleus of an atom.
- b** Why are the electrons shown as circuits around the nucleus in diagram 1?
- c** Diagram 2 is a type of model used regularly in science books where the electrons are arranged in electron 'shells' around the nucleus. What is wrong with using the analogy of 'shells' in this description?
- 2.2** Diagrams or clusters of atoms (molecules) are also common in science texts. Study the model made of balls and rods in diagram 3. This model represents a molecule of methane which is made up of one carbon atom and four hydrogen atoms.

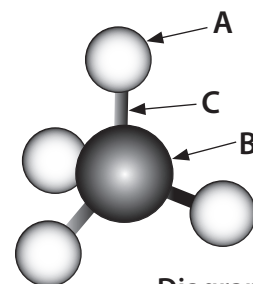
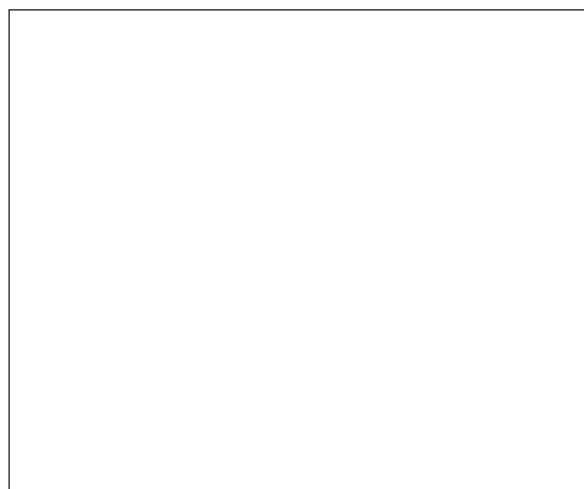


Diagram 3

- a** What does 'A' represent?
- b** What does 'B' represent?
- c** What does 'C' represent?
- 2.3** For the purpose of discussion, a ball is often used to represent the Earth and a torch to represent the Sun. This model allows us to 'view' the Earth – we can rotate the ball and think about night and day, the seasons and Earth's orbit around the Sun.
- a** Draw and annotate a diagram in the box at the right to show how this model works.
- b** Like most models, this model has limitations. List two of these limitations.
- 2.4** A model may have limited usefulness because it is just too simple. Because molecules are too small to see, we are often asked to 'imagine' that a gas is like a lot of tiny balls that are endlessly moving about, bouncing off one another. What is one thing that this mental picture is unable to account for?



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.



From a scientist's point of view

Scientists share certain attitudes and ways of thinking that are related to the nature of the world and what can be learned about it. As our understanding of the world evolves, scientists are always prepared to alter their opinions and theories to adapt to new evidence.

Activity 1

1.1 The table below lists ten attitudes that scientists possess. Draw lines to match each attitude with how it relates to scientific enterprise.

SCIENTIFIC ATTITUDE		HOW THE ATTITUDE RELATES TO SCIENTIFIC ENTERPRISE	
a	Thirst for understanding and knowledge	• Seeing new possibilities in the data	
b	Being sceptical	• There is no absolute certainty in knowledge	
c	Relying on data	• Valuing patterns of thought that lead from data to conclusions to the development of theories	
d	Accepting ambiguity	• Nature's relationships are sometimes best revealed by patterns and formulas.	
e	Willingness to modify explanations	• Scientists are addicted puzzle-solvers	
f	Sharing knowledge	• Viewing information objectively and without bias	
g	Respecting reason	• Explaining natural occurrences by collecting and ordering information, testing ideas and respecting the information that is revealed	
h	Being honest	• Collaborating and co-operating in answering questions and solving problems – working together to pool ideas, explanations and solutions	
i	Understanding that all knowledge has tolerance limits	• Recognising that there are appropriate times and places to question authoritarian statements and obvious 'truths'	
j	Respect for mathematics as a language of science	• Recognising that data is rarely clear; appreciating that new questions and problems do arise	

Breaking the rules

Our knowledge constantly increases and evolves as scientists discover more about the world around us. In all kinds of science, explanations are constantly put to the test. If an experiment shows that a rule can be broken, the rule has to be changed. Occasionally, an important new discovery revolutionises a whole branch of science.

Activity 2

- 2.1** In 1826, the German chemist Fredrich Wohler heated some chemicals and, quite by accident, made a substance called urea. At that time, urea could only be made by animals. Wohler's discovery helped to disprove the theory that the chemistry of living things was different to the chemistry of non-living things. Scientists came to realise that the two kinds of chemistry worked in the same way.
- a After Wohler's discovery, what did scientists believe about the chemistry of living and non-living things?
 - b What 'old rule' had been replaced?
 - c Madame Curie discovered by accident that minerals containing uranium darkened photographic paper. What did Wohler discover by accident?

2.2 A valuable spin-off of science investigation is that of complementary knowledge, where links between unrelated pieces of information are identified. A molecular biologist saw the units of inheritance called genes. Then a chemist found out about the chemical structure of DNA. Now geneticists are searching for the codes that cause genetic diseases and zoologists are applying this knowledge to unravelling evolution.

- a** Draw a mind map of complementary knowledge. Start with 'discovery of genes' in the middle, and add the other scientific discoveries and research from the information above.
- b** Then think of 10 more 'spin-offs' that have occurred from the discovery of genes and add them to your mind map.
- c** Finally, add a further 10 spin-offs that you think will happen in the next 20 years.

Radical ideas in science

All sorts of people from famous scientists to everyday people have contributed to the accumulation of scientific knowledge over many centuries. This progress has generally been gradual, but from time to time, radical 'leaps' in knowledge occur. One example is Galileo's contribution to changing our perception of our place in space.

500 years ago, people believed that the Earth did not move. It was believed to be the centre of the universe, which everything moved around. The Sun and the stars rose in the east and set in the west. The people of Aristotle's society were quite comfortable with this theory that the Sun and stars were all fixed to a giant sphere that moved around the Earth.

In 1542, a Polish astronomer named Copernicus put forward the idea that the Sun was the centre of the universe, not the Earth. Copernicus knew that this view contradicted commonly held ideas of the time, and his ideas were rejected by almost everyone.

Decades later, Galileo heard of Copernicus' ideas. Galileo believed these ideas as they supported some observations of his own that he had made with one of the first telescopes. When he looked at Jupiter, he found that there were three stars that appeared to move around Jupiter over several days. He thought that these must be moons of Jupiter.

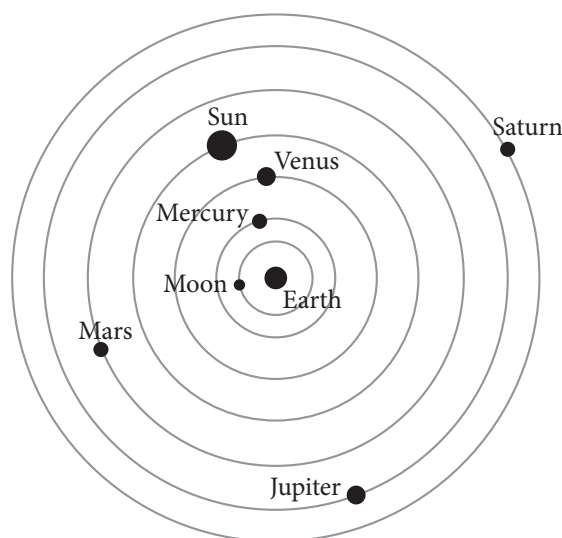
After thinking some more, Galileo argued that if Jupiter had moons moving around it, then not everything in the universe moved around the Earth, and just maybe it was the Earth that moved around the Sun. He presented sound arguments in his writings and hence drew the attention of the educated people of the time. This theory created political, religious and scientific controversy.

Activity 3

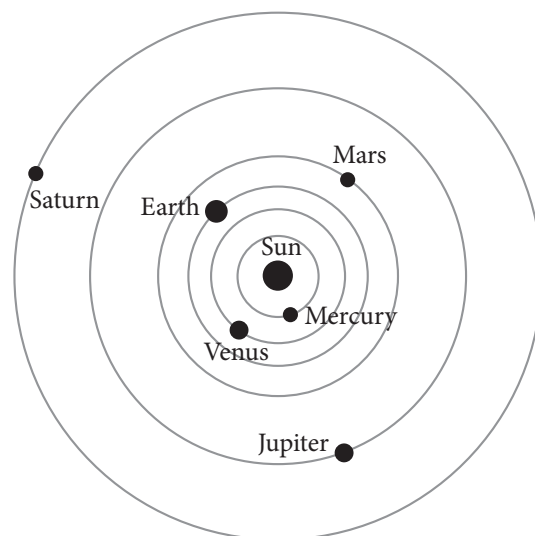
3.1 Use the information above to answer the following questions.

- a** What was Aristotle's theory about the Earth's place in space?
- b** What was the Copernican theory about the Earth's place in space?
- c** What observation did Galileo make which supported the Copernican theory?
- d** What was Galileo's hypothesis about the movement of the Earth and the Sun?
- e** What new piece of technology did Galileo have that Copernicus did not?
- f** Galileo did not have any equipment for conducting experiments. How then did he try to convince others of his ideas?
- g** What made Galileo confident that his ideas were correct?

The Earth-centred model



The Copernicus model



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

Spreading the word

When people know how scientists go about their work and reach scientific conclusions and what the limitations are of such scientific conclusions, they are more likely to react thoughtfully to scientific claims and less likely to reject them completely or accept them uncritically.

Myth and superstition

The images that many people have of science and how it works are often distorted. Myths and stereotypes about the role of scientists in our daily life are plentiful and can interfere with our understanding of science and our ability to analyse and make decisions about its findings.



Activity 1

1.1 Study the stereotype of the scientist.

- a List five common misunderstandings about scientists and their work.
- b How do these negative stereotypes affect our ability to see science as a part of the everyday world?

1.2 A concept map or spider diagram is a method of exploring links between ideas on a topic. How do you 'see' science? On a blank piece of paper, draw a concept map of your understanding about the nature of science. Some words are provided below to help you get started.

Tips: Put the most inclusive word in the middle of the page. Write the other words around it and connect the words with lines on which you describe the connection between the words.

For example:



- science
 - scientists
 - general public
 - media
 - experimenting
- validating
 - observing
 - thinking
 - knowledge
 - intellect
- understand the world
 - instruments
 - patterns in nature
 - invent theories
- test ideas
 - find answers
 - ethics
 - consider evidence
 - make predictions
- communication
 - make decisions
 - recognise limitations
 - considered opinion

Scientists talking to scientists

Scientists talk to each other at conferences, in science meetings and through email. They publish in scientific journals and publications. Clear communication enables scientists to inform other scientists about their work, expose their ideas to criticism and stay informed about scientific discoveries around the world.

Activity 2

2.1 When scientists talk among themselves, they use technical language, specialised terms and specific jargon. Write these scientific terms beside the correct 'everyday' explanations in the table: *epidemic, habitat, emission, greenhouse effect, extinction, global warming, producer, gene, virus, clone*.

SCIENTIFIC WORD	'EVERYDAY' EXPLANATION
a	The place where an animal or plant lives.
b	Genetically identical individuals or cells.
c	A living thing that uses the Sun's energy to turn simple substances into food.
d	An increase in the Earth's average temperature.
e	Expulsion of a used product.
f	A disease prevalent in a community at a particular time.

g	The trapping of heat by the Earth’s atmosphere.
h	A basic unit of heredity that carries the instructions needed to make a particular protein.
i	The permanent disappearance of a species.
j	Disease-causing agents, smaller than bacteria, that only grow inside a cell.

Technical jargon

Technical jargon tends to interfere with communication between scientists and the general public. Science writers for newspapers, magazines and television try to translate the highly technical language of science into language that is accessible to an average person. To do this, they assume that the person is familiar with the main ideas of science and is therefore able to process this information.

Activity 3

Plant industry researchers have harvested the first plot of transgenic wheat ever to be grown in Australia. The wheat contains genes from another grass variety that will increase the amount of wheatflour that is suitable for noodle making. The research was carried out under the guidelines of the Genetic Manipulation Advisory Council. The researchers said that other improvements such as disease resistance and grain quality may also be made, but that it will be several years before these wheats are available to the farmers.

3.1 Use the information above to answer the following questions.

- a What is transgenic wheat?
- b What was the purpose of this research?
- c Which organisation provided some control over this research?
- d What other improvements are the researchers hoping to achieve?
- e Why will it be several years before the improved grain variety is available to farmers?

Science solving problems

Scientists can bring information, insights and analytical skills to bear on matters of public concern. Acting in their areas of expertise, scientists can help people to understand the likely causes of events and estimate their possible effects.

Activity 4

It has been reported that many of Australia’s rural and urban areas are experiencing less rainfall and, consequently, water shortages. Four scientists were consulted and their comments are below.			
Scientist 1 – The climate models predict that in the future, rainfall will be more variable.	Scientist 2 – The climate is changing and it would be prudent for us to assume we are going to get less rainfall.	Scientist 3 – There is considerable uncertainty about how Australia’s future average rainfall might change.	Scientist 4 – All the new sources of water such as desalination plants are more expensive and more energy intensive.

4.1 Use the information above to answer the following questions.

- a Can scientists prove that the changing rainfall pattern is due to climate shift?
- b Can scientists be sure about how the rainfall may change?
- c From what the first three scientists said, can the general public conclude that climate change is likely to alter rainfall patterns and that we should become more water wise?

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.



Code of Ethics

There are accepted ethical principles in the conduct of science. Most scientists follow strongly held traditions about accurate record-keeping, openness and replication of results, supported by the critical review of their work by peers.

d What does Scientist 4 warn about?

Look out for scientific dishonesty

Responsible scientists are conscientious scientists – they care about procedures that protect both the integrity of science and its reputation.

The strongly held traditions of science, including its commitment to peer review and publication, keep the majority of scientists well within the bounds of ethical professional behaviour. Deliberate deceit is rare and likely to be discovered and exposed sooner or later by the scientific community.

However, sometimes the pressure to get credit for being the first one to publish findings or to get recognition for a discovery leads some scientists to withhold information or even falsify their findings. When violations of these scientific ethical traditions are discovered, they are strongly condemned by the scientific community. The violators can lose their jobs, face criminal charges and be barred for life from receiving funds to continue research. Also, it is extremely unlikely that they will ever regain the respect of other scientists.

Scientific misconduct wastes time and money – but more importantly, it can be a danger to the public as well as damage the reputation of science and scientists.

Activity 1

1.1 Which of the following are examples of scientific misconduct?

- a** Academic plagiarism – stealing another scientist's results and claiming them as your own
- b** Providing false information when applying for research funding
- c** Falsifying data
- d** Exaggerating the number of subjects in an investigation
- e** Bypassing peer review – publishing in the media instead of in an accepted scientific journal
- f** Repeating another scientist's work in order to verify the results

1.2 List three possible consequences of scientific misconduct for:

- a** the scientist found guilty of misconduct.
- b** the scientific area in which that scientist worked.
- c** the public.

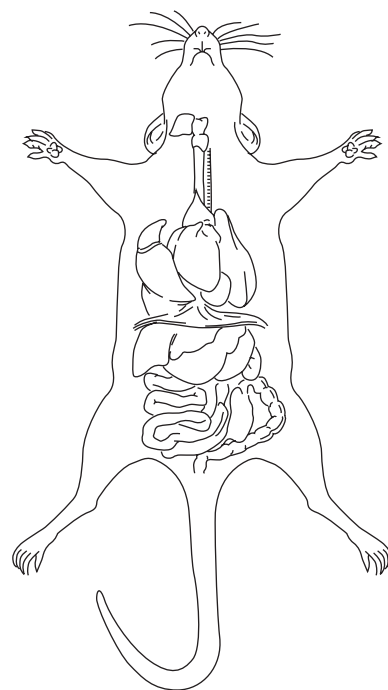
Deliberations about live animals

One area of scientific ethics is the treatment of animals – in particular, any harm caused to animals by scientific experiments. Research using animals is governed by strict guidelines regarding health, comfort and wellbeing.

Scientific and medical researchers routinely use animals in experiments. Even today, these experiments often expose animals to less than ideal circumstances. The manufacturers of drugs, cosmetics and household products as well as those involved in medical advances and veterinary treatments still use animals in research. According to the Australian Association for Humane Research, more than 6 million animals were used in research and teaching in Australia in the year 2006.

(Source: www.aahr.asn.au/statistics.html)

What do you think? Should we allow animals to be used in research?



Activity 2

2.1 For each product in the table below, decide whether you think animal testing should be allowed. Assume that the animals will experience a medium level of discomfort, but with no long term ill effects. Give reasons for your answers.

PRODUCT	ALLOW ANIMAL TESTING? WHY?
A new dishwashing detergent that leaves the user's hands soft and moisturised	
A sunscreen that provides a higher level of protection from UV radiation	
A cloning technique that will provide a way for scientists to test products on cloned cells instead of on animals	
A cure for Alzheimer's disease	
A cure for cancer	

- 2.2 Consider your answers to the previous question. If for each product, the animals tested will experience a high level of discomfort resulting in death, would this change your decisions? Explain your answer.
- 2.3 Should all living things – for example, bacteria, worms, rodents, insects, fetuses and people in irreversible comas – have equal rights when it comes to scientific research? Explain your answer.

Tips for making difficult decisions

- Be able to distinguish between technical and ethical issues.
- Recognise that innovations may have both benefits and risks, including the unforeseen.
- Appreciate that decisions about solutions to problems are influenced by things like technical feasibility, ethical implications, economic cost and social and environmental impact.

Activity 3

REPORT ON CHEMICALS IN PACIFIERS (DUMMIES) FOR INFANTS

Chemicals called phthalates are added in very small amounts to PVC plastic to make it soft enough to form flexible shapes. In very large amounts, phthalates can cause liver damage and may cause cancer. Pacifiers and other baby products are made of this plastic.

Some scientists are concerned about the level of phthales in PVC plastics. They tested the level of phthalates in several types of pacifiers by shaking the pacifiers in artificial saliva to mimic what might happen in a baby's mouth. They found that three brands of pacifiers released large amounts.

Maximum level of phthalate allowed in food (micrograms)	50
Dummy 1	2200
Dummy 2	1000
Dummy 3	9

- 3.1 The scientists think these pacifiers should be withdrawn from sale. They have no information about the effect these pacifiers have on infants using them. Which of the following issues is the most important one to consider in deciding whether to put phthalates in PVC plastic used for baby products? Explain your answer fully, using as much information from the report as you can.
- a The plastic must be soft enough to form products suitable for babies.
 - b There must not be any chemicals in the plastic that may cause harm.
 - c The risk of damage to babies from using softened plastic has to be compared with the risk from chemicals causing harm.

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.



The Scientific Method – an everyday tool

The scientific method is simple and makes really good sense. Some people make a lifetime career out of defining scientific method – many other people just use it!

The scientific method can be applied to many everyday situations, decisions and choices. It's all about examining the problem, proposing a solution and then putting that solution to the test. Of course, proposing the solution is difficult, requiring imagination, knowledge and self criticism – but testing that solution is the important part.

From theory to law

In science, a **law** is a statement or explanation that always appears to be true. A scientific law or principle is a **theory** that has been thoroughly tested and shown to be true. For example, Bernoulli's Principle and Boyle's Law are scientific laws about the pressure of a gas.

Investigating Murphy's Law

Murphy's Law: Whatever *can* go wrong *will* go wrong

Murphy's Law is not a scientific law, but an 'adage'. An adage is a commonly known saying that many people consider to be true. Often they are pessimistic, like Murphy's Law. 'Don't judge a book by its cover', 'practice makes perfect' and 'you can't have your cake and eat it too' are also common adages.

One well known example of Murphy's Law is 'Murphy's Law of Toast'. This 'law' states that if toast can land butter side down, making the biggest possible mess, it will do just that. Scientists have conducted controlled experiments to test this 'law'.

Activity 1

1.1 Consider the information in the box below. Then answer the questions on the next page to help you design and conduct your own experiment to test Murphy's Law of Toast.

- ▲ The centre of gravity of the toast is altered by the weight of butter on one side and not the other.
- ▲ A piece of buttered toast loses contact with the plate immediately after its centre of gravity passes over the edge of a plate.
- ▲ Physics and gravity dictate that toast can only turn by 180° if it falls from a height of 1 metre. Theory suggests that toast falling from 3 metres will always land butter-side up.
- ▲ Trials need to be large enough to gather enough evidence to reach a result that is unlikely to be called a 'fluke' and give a reasonable chance of revealing a genuine effect if one exists.
- ▲ If there is bias towards butter-side down landings, then it is reasonable to assume that this bias will reveal itself after 20 trials.
- ▲ The more times a trial is repeated, the more confidence you can have in the results. It is reasonable to assume confidence in the results if 50 independent trials were conducted.
- ▲ Some published results show that from a height of 1 metre, a butter-side up result occurs 61% of the time with buttered toast. With unbuttered toast, the side facing up at the time of the fall lands facing up 59% of the time. From a height of 3 metres, butter-side down occurred 49% of the time.

- a** What do we call a 'law' before it has been adequately tested?
- b** What question are you investigating?
- c** What is your hypothesis?
- d** If further experiments continue to support a hypothesis, it is then called which of the following?
- ▲ data
 - ▲ a theory
 - ▲ an accepted fact
 - ▲ conclusive evidence
- e** What equipment will you use to conduct this investigation?
- f** In your opinion, would the following suggested trial size and method provide sufficient confidence in the results?
- Place normally buttered toast on plates held at chest height (1 metre). Then tilt the plates until the toast slides off and lands on the floor. This procedure is to be repeated 20 times each by 50 students in each of 50 different schools.*
- g** Which of the following should a properly conducted scientific experiment **not** have?
- ▲ more than one variable
 - ▲ close observation
 - ▲ accurate data
 - ▲ more than one control
- h** What safety issues should be considered while conducting this experiment?
- i** Murphy's Law can be applied to many situations. For example, Murphy's Law of Sticky Tape states that 'the more of a rush you are in, the harder it is to find the start of the roll of tape'. Is there a scientific basis to statements like these that involve Murphy's Law?
- j** This investigation has been done for the sake of knowledge alone. What is this type of research called?
- ▲ applied research
 - ▲ experimental science
 - ▲ applied science
 - ▲ pure science
- k** If a 'law' of science is a statement or explanation that always appears to be true, then what is it that may be causing bias in the community's perception of the validity of Murphy's Law of Toast?
- l** A second experiment in which all factors are the same except the one to be tested is referred to as which of the following?
- ▲ a control
 - ▲ a variation
 - ▲ an experimental factor
- m** How could you control for any bias caused by the weight of the butter on one side?
- n** What is the one differing factor between a control group and an experimental group?
- ▲ the trial answer
 - ▲ the theory
 - ▲ a variable
 - ▲ a hypothesis
- o** A working explanation or trial answer is called which of the following?
- ▲ a theory
 - ▲ the trial answer
 - ▲ a variable
 - ▲ a hypothesis
- p** How could you investigate the claim that if buttered toast was dropped from 3 metres it would always land butter-side up?
- q** Is butter the cause of butter-side down results?
- r** Is the height the cause of butter-side down landings?
- s** How do the results for the 3 metre drop support what physics and gravity have to say?
- t** What conclusion can you draw from your results?
- u** Scientists believe that butter-side down landings are no more likely than butter-side up landings. Do your results show this?



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

1 – Is seeing believing?

- 2.1 Open questions: *c, e*. Closed questions: *a, b, d*.
 2.2 Students' questions could relate to topics such as global warming, possible life on meteorites, cancer risk from stem cells, dark matter, superconductivity and avian influenza.
 2.3 *c* For example: Is its skull large or small? Is its neck long or short? Is its tail for balancing? Does it have tall bones from its backbone?
 4.1 Jake was a cat and Rosie was a bird.
 4.2 The man was bald.

2 – Give it a name!

- 1.1 From mother: hair colour. From father: eye lashes, dimples, ability to raise one eyebrow only.
 2.1 A: Rock 3. B: Rocks 2, 3, 5. C: Rock 1.
 3.1 Plants: B, D. Animals that live in water: G. Animals that do not live in water: A, C, E, F, H, I. Animals with bones: A, C, E, H, I. Animals without bones: F, G. Animals that are eaten by other animals: A, E, F, G, H, I. Animals that are not eaten by other animals: C.
 3.2 Easily seen: Sharks – rough skin and a tail with one paddle; Dolphins – smooth skin and a tail with two paddles. Not easily seen: Sharks – breathe with gills, flipper has stiff rods made of cartilage; Dolphins – breathe with lungs, flippers have bones arranged in rows like finger bones.
 4.1 Genus name: *Macropus*; Species name: *rufus*; Common name: red kangaroo.
 4.2 By grouping things which are similar, we can better understand the relationships between objects, between living organisms, and between living and non-living things.

3 – Dreaming up solutions

- 1.1 *a* Traffic congestion, ride a bike or a bus. *b* Borrow the book from a friend nearby.
 1.2 *a* Global warming, weather patterns *b* Reduce fossil fuel use, reduce methane production by animals and rubbish dumps.
 2.1 *a* Why isn't the torch working? *b* Dead battery, a broken bulb, a corroded contact.
 2.2 Dissolve the salt in water, strain the solution to retrieve the pepper and then evaporate the water from the salt solution.
 5.1 *a* No *b* The shell had been removed and the egg felt like rubber and was called 'rubber', but it was just a raw egg without its shell so of course it burst.
 6.1 *a* Take a straw and position one end of it level with the space between the balls and then blow firmly – the balls will swing together. *b* The pressure of the moving air between the balls was less so the higher pressure of the surrounding air forced the balls together.
 6.2 If the kite is designed correctly, the kite meets the air with a greater pressure on its front than on its back which creates lift.
 6.3 The petrol engine was able to drive the rotor blades fast enough to generate lift.
 7.1 *a* The newspaper is probably not strong enough to hold the weight of the potatoes. *b* It would tear and spill the potatoes. *c* Fold the paper to make it stronger, use more sticky tape at weak points.

4 – Design – from concept to construction

- 2.1 Pure research: *a, f, h, i, j, k, l, m, n*. Applied research: *b, c, d, e, g, o*. Knowledge of all of this information is useful.
 3.1 *a* The most fluid polymer might be used for the sole under the flexing part of the foot – the midsole. *b* The stronger polymers would be appropriate for the part of the sole that protects the feet from the impact with the ground – the heel. *c* The sole protecting the ball of the foot. *d* Test the materials for ability to withstand repeated impact. *e* For example, *d* – it does not matter how the polymer used to make the sole of a shoe feels against the skin.

5 – What is a good hypothesis?

- 1.1 *a* UV radiation *b* Does wearing a T-shirt protect you from UV radiation when you are swimming?
 2.1 1 An interesting observation is made by a researcher. 2 The researcher asks questions in order to explain the observation. 3 A hypothesis or educated guess is devised as the answer. 4 An experiment is designed to test the hypothesis. 5 The experiment is conducted. 6 The hypothesis is accepted or not accepted.
 2.2 True
 3.1 *a* Skin often becomes red and sore if exposed to sunlight. *b* Sunlight damages the skin. *c* Students may have first-hand experience with sunburn. *d* Over-exposure may cause serious and long-lasting skin damage.
 4.1 *b*
 4.2 *b*
 4.3 *a*
 4.4 *b*
 4.5 *a*
 4.6 *a* Wearing a t-shirt protects you from UV radiation while you are swimming.

6 – Determining the variables

- 1.1 *a* Brand of chocolate (production process), type of chocolate (Is it for cooking? Is it cheap or expensive?), melting temperature.
 1.2 *a* Brand, type of chocolate (milk) *b* Taste *c* Different groups of people have different preferences, according to factors such as age, sex and ethnic background. *d* They could rate each chocolate from 1 to 4. *e* Possibly.
 2.1 The manipulated variable is the presence of cocoa butter, the responding (dependent) variable is the type of chocolate (milk, dark, cooking) and the fixed variables (controlled) are the brand and the temperature.
 2.2 Control variable – seed, water, light, temperature
 Manipulated variable – salt level of the soil
 Responding variable – plant growth
 3.1 *a* Water temperature, dissolving speed, e.g. amount of salt, type of salt, rate of stirring *b* Hardness of sporting ball, height of bounce, e.g. size of the balls, bouncing surface, height of ball drop *c* Tap design, size of the drip, e.g. rate of water flow, material the tap is made from, temperature of the water *d* Birth date, weight of people, e.g. age, sex, ethnic background, height, eating habits

7 – Great experimental design

- 1.1 Step D, Step C, Step B
 2.1 Step Y, step V, step X, step Z, step W
 3.1 *a* Mice do not die if given human antibodies. *b* Yeast cells contain human antibody DNA. *c* Human DNA can make human antibodies. *d* Mice can make human antibodies.
 4.1 *c, b, e, a, d*
 4.2 Independent variables: brand of toilet paper. Dependent variables: fragmentation. Control variables: weight of toilet paper, amount and temperature of water, degree of shaking.
 4.3 Yes
 4.4 Yes
 4.5 Materials biodegradability is assisted by a larger surface area.
 4.6 How much fragmentation there is.
 4.7 No – it is a subjective measurement. For example, a grading system of 1 to 10 (where 10 is most fragmentation) could be used.
 4.8 Yes

8 – Measuring up

- 1.1 Sight, hearing.
 1.2 Photoelectric cells, movement, set off a burglar alarm smoke detectors, smoke, set off fire-extinguishing systems thermostats, changes in temperature, keep air conditioners at the required temperature.

- 2.1** **a** Beaker – holding liquids **b** Filter funnel – separating solids and liquids **c** Gas jar – collecting and storing a gas **d** Pipette – dispensing an accurate volume of liquid **e** Tap funnel – adding a controlled volume of a liquid **f** Burette – adding accurate volumes of liquid **g** Conical flask – holding liquids **h** Bunsen burner – supplying heat **i** Tripod – as a stand for flasks and beakers during heating **j** Round-bottomed flask – heating liquids
- 3.1** **a** Volume **b** Air pressure **c** Distance **d** Time **e** Weight

9 – Number crunching

1.1

Quantity	Unit of measurement	Abbreviation	Relationship between units
Mass	kilogram gram	kg g	1000 g = 1 kg
Area	square metre square centimetre	m ² cm ²	1 m ² = 10 000 cm ²
Energy	joule kilojoule	J kJ	1000 J = 1 kJ
Temp.	degrees Celsius	°C	(n/a)
Time	second minute	s min	60 s = 1 min

- 1.2** **a** Millimetre (mm) **b** Kilogram (kg) **c** Square kilometres (km²) **d** Litres (L) **e** Light years
- 2.1** To simplify and enable communication and commercial transactions between different nations and cultures.
- 2.2** **a** 60 cubic centimetres **b** Volume = length x breadth x height
- 2.3** **a** 2.5 g/cm³ **b** Grams per centimetre cubed (g/cm³) **c** 0.025 kg/m³
- 3.1** 120 kPa ± 10 kPa
- 3.2** **b** 2.10 s **c** 0.21 s **d** Highest – 0.25 s, lowest 0.17 s **e** 0.04 s **f** 0.04 s **g** 0.04 s **h** 0.21 s ± 0.04 s
- 3.3** 17 ± 4

10 – In the wild – the ultimate laboratory

- 1.1** **a** 3 km @ 5 km/h + 7 km @ 3 km/h + 2.5 km @ 1.5 km/h = 4.6 hours **b** 5 km @ 0.5 km/h = 10 hours

1.2

Altitude (m)	Vegetation on south-facing slopes
550–150	Sparse tree cover
150–100	Sparse shrub cover
100–50	Thick bush
50–0	Weed patches

- 1.3** AC
- 1.4** **a** Site B **b** The characteristics indicate the overall health of the plant.
- 1.5** The plants were already unhealthy, insects like warm temperatures and low winds.
- 1.6** Between 6 and 10 kilometres

11 – A picture tells a thousand words

- 1.1** **a** The flask should be flat-bottomed. **b** The stopper has gaps down the side. **c** The lines on the stopper appear to cut through the stem of the funnel. **d** The end of the funnel should be under the liquid, otherwise the gas would escape. **e** The tube appears to be breaking through the beehive shelf.
- 1.2** **a** Wheel Y will go anticlockwise and wheel Z will go clockwise. **b** Wheel Y **c** To show movement in a particular direction **d** Three, wheel Z, twice, wheel Y.
- 2.1** **a** Clockwise **b** From west to east **c** 46 km **d** Destructive **e** × **f** A dotted line
- 3.1** **a** Sub-categories **b** Energy transfer – the organisms or creatures that are eaten by other organisms or creatures **c** Direction of energy transfer
- 4.1** **a** 7 metres **b** 1 cm represents 1 m **c** Sailboat = 6.5 metres, balloon = 3 metres **d** 32.5 metres

12 – Integrating information

- 1.1** **a** Discrete **b** Continuous **c** Discrete **d** Continuous **e** Discrete
- 1.2** Pie chart – discrete – D. Histogram – continuous – B. Bar graph – discrete – A. Scatter graph – continuous – E. Line graph – continuous – F. Column graph – discrete – C
- 2.1** **a** Rules 1, 2, 3, 4. **b** Rules 1, 4, 5.
- 2.2** **a** Pie chart **b** Line graph **c** Bar or column graph **d** Histogram **e** Histogram

13 – What does it all mean?

- 1.1** **a** Seagull **b** Bread **c** London
- 1.2** e
- 1.3** Z X Y. The pattern is an X followed by YY and another X. The pattern repeats.
- 2.1** **a** Aluminum **b** 4 km **c** Aluminium **d** Site B: A car yard, as the damage to iron is least. Site X: A marina, as the stainless steel is less affected than the other two metals at an acid level of 0.8 µg.
- 2.2** **a** This line is roughly parallel to the horizontal axis so it represents no change in the metal as the result of acid levels. **b** The aluminum line demonstrates an almost constant rate of damage. **c** 0.5–0.7 µg of acid as this is where the line is steepest. **d** Extend the last segment of the graph to 1.0 µg/m³ of acid and read the damage from the y-axis = 68 cm/m³.
- 3.1** **a** Method is suitable but would not provide enough valid data. **b** Accuracy and reliability are acceptable but only for the scope of this study. **c** Use more sets of metal plates at regular distances from the factory. **d** Level of acid does not affect stainless steel, but corrosion of the other metals was fairly constant as acid level rose. **e** Damage to the major metals involved in the developments increases (is directly proportional) with the level of acid in the air except for stainless steel. Acid levels in the air decrease as distance from the factory increases (indirectly proportional). **f** Not really, as there were not enough replicates of the metal plates and so trends only could be assessed. A more comprehensive study would need to be done before money was invested in these developments. Also, the prevailing winds were not considered. **g** Place plates at more regular intervals and in a circle around the factory, take into account the prevailing winds and increase the length of the study.

14 – Learning from each other

- 1.1** To make it easier for a variety of scientists to read and understand.
- 1.2** [In order] literature review, aim, findings, method, contribution to knowledge
- 1.3** **a** How do I describe the content of this report in the smallest number of words? **b** Why am I doing it? **c** What do we know? What don't we know? **d** What do I hope to discover? **e** How am I going to discover it and what equipment do I need? **f** What have I found out? **g** What does it all mean? **h** What did I find out? What are the possible applications and recommendations? Do the results support the hypothesis or not? What next?
- 2.1** **a–c** – see answer to 2.2 below. **d** pH meter and GPS. **e** B
- 2.2** Revised version of report:
The research team was interested in what plant species were growing where the pH of the soil had been increased by alkaline runoff from a building site. The equipment required is listed below. The following data was recorded – exact location of each quadrat, the plant species and the pH in each quadrat. After analysing the data, the team concluded that the high pH levels stopped the growth of only some plants.
- 15 – It's all about common sense**
- 1.1** **a** Rubber gloves **b** Splash goggles **c** Face masks and respirators **d** Safety glasses **e** Laboratory coats **f** Safety helmets and visors
- 2.1** Spatulas, tongs
- 2.2** Fume cupboard
- 2.3** Safety shower, eyewash station

3.1

Type of extinguisher	Use	Part of the fire triangle removed
Sand bucket	Small fires	Oxygen
Fire blanket	Fires on people	Oxygen
Fire extinguisher	Larger fires where the fuel is known	Oxygen

- 4.1 **a** Breathing in fungal spores or bacterial cells **b** Poisonous plants
c Infections with parasites **d** Coming into contact with animal-produced toxins.
- 6.1 **a** Put rubbish in the correct bin. **b** Flush only allowed chemicals down the sinks, with plenty of water. **c** Do not use containers provided for other chemicals. **d** Use just enough chemical for the procedure so that there is nothing left over for disposal.

16 – Understanding models

- 1.1 **a** So that the angle of the wing can be adjusted to the air flow.
b How much weight the wing can carry. **c** It would not lift as well. **d** The cotton would show how the air particles 'flow' over the wing. **e** The wing would slow down (drag).
- 1.2 **a** e.g. cycling, running **b** e.g. bike helmets, clothing
- 1.3 So that any adverse results will be discovered before its use on people.
- 1.4 These organisms have large numbers of generations in a relatively short time span.
- 1.5 1.3: Medical treatments may have different consequences when used on people. 1.4: Genetic transfer in flies, for instance, may be completely different to what happens in humans.
- 2.1 **a** Protons and neutrons **b** To indicate that electrons have enormous amounts of energy and move around the nucleus.
c Shells have hard surfaces and electron pathways do not.
- 2.2 **a** One atom of hydrogen **b** One atom of carbon **c** The chemical bond between hydrogen and oxygen
- 2.3 **b** The Earth is not perfectly round and cannot be bounced, the torch is the wrong shape for the Sun.
- 2.4 This mental picture does not account for the moving parts inside the balls.

17 – Scientific attitudes

- 1.1 **a** Scientists are addicted puzzle-solvers. **b** Recognising that there are appropriate times and places to question authoritarian statements and obvious 'truths'. **c** Explaining natural occurrences by collecting and ordering information, testing ideas and respecting the information that is revealed.
d Recognising that data is rarely clear; appreciating that new questions and problems do arise. **e** Seeing new possibilities in the data. **f** Collaborating and cooperating in answering questions and solving problems – working together to pool ideas, explanations and solutions. **g** Valuing patterns of thought that lead from data to conclusions to the development of theories.
h Viewing information objectively and without bias. **i** There is no absolute certainty in knowledge. **j** Nature's relationships are sometimes best revealed by patterns and formulas.
- 2.1 **a** The chemistry of living and non-living things was the same.
b The chemistry of living and non-living things was different.
c A substance called urea, that previously could only be made by animals.
- 3.1 **a** Earth was stationary and the Sun and stars revolved around it.
b Copernicus believed that the Earth revolved around the Sun as well as on its own axis. **c** Stars moving around Jupiter.
d Earth moves around the Sun. **e** A telescope **f** Putting forward arguments in common language, indicating the evidence that he did have. **g** His observations and evidence fitted with the observations and evidence of Copernicus.

18 – Are you scientifically aware?

- 1.1 **a** For example: male, old, balding, wears glasses, wears a white coat, works in a place with weird equipment, conducts mysterious bubbling experiments, looks like a nerd. Science is carried out by strange people, who work in isolated places, do strange things and talk in strange ways about things that are remote to our everyday experiences. **b** People can believe that science is exclusive and difficult, rather than realising that science is all around us, and even 'regular' people can observe and use science in everyday life.
- 2.1 **a** Habitat **b** Clone **c** Producer **d** Global warming **e** Emission
f Epidemic **g** Greenhouse effect **h** Gene **i** Extinction **j** Virus
- 3.1 **a** Wheat into which a gene from another type of plant has been substituted. **b** To produce wheat high in the flour used to make noodles. **c** The Genetic Manipulation Advisory Council.
d Disease resistance and higher quality grain. **e** There is much more research to be done to make sure that the genetically engineered crop is safe for humans and the environment.
- 4.1 **a** No **b** No **c** Yes **d** The public will pay more for water in the future.

19 – Ethics in science

- 1.1 a, b, c, d, e
- 1.2 **a** Lose job, face criminal charges, loss of respect of other scientists, barred for life from receiving funds to continue research, wasted use of time and money and the damage to the reputation of scientists. **b** Damage to the reputation of scientists, misleading results may invalidate other research conducted in the same area. **c** Potential physical danger to the public, damages public opinion of scientists in general.
- 3.1 **a** This option shows evidence for one side of the issue only.
b This option shows evidence for both sides of the issue. **c** This option shows evidence on both sides of the issue and indicates how the balance should be decided.

20 – Don't believe everything you hear!

- 1.1 **a** A theory **b** Why does toast always land butter-side up?
c Toast is as likely to land butter-side up as butter-side down.
d A theory **e** Toast, butter, a knife and a plate **f** Yes **g** More than one variable **h** Researchers may slip on a buttery floor.
i Not really – they are referred to as 'urban myths'. **j** Pure science
k Selective memory – they only remember the times that the butter made a mess and not the times when the toast landed cleanly. **l** A control **m** Repeat the investigation using toast with a 'b' for butter written on the upper side of the toast. **n** Variable
o A theory **p** Repeat the two experiments but drop the toast from a height of 3 metres. **q** It must, as 59% is 9% higher than scientists believe. **r** The 61% rate found for normal buttered toast appears to confirm Murphy's Law of Toast in that there is a bias towards butter-side down landings, however the close results obtained for buttered (61%) and unbuttered toast (59%) does not make this very convincing. Also, the sub-50% result for the 3 metre toast would indicate that height does have an effect.
s That toast must have enough distance to fall 360 degrees so that it can turn right over and land right-side-up.

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